

Predictive Maintenance

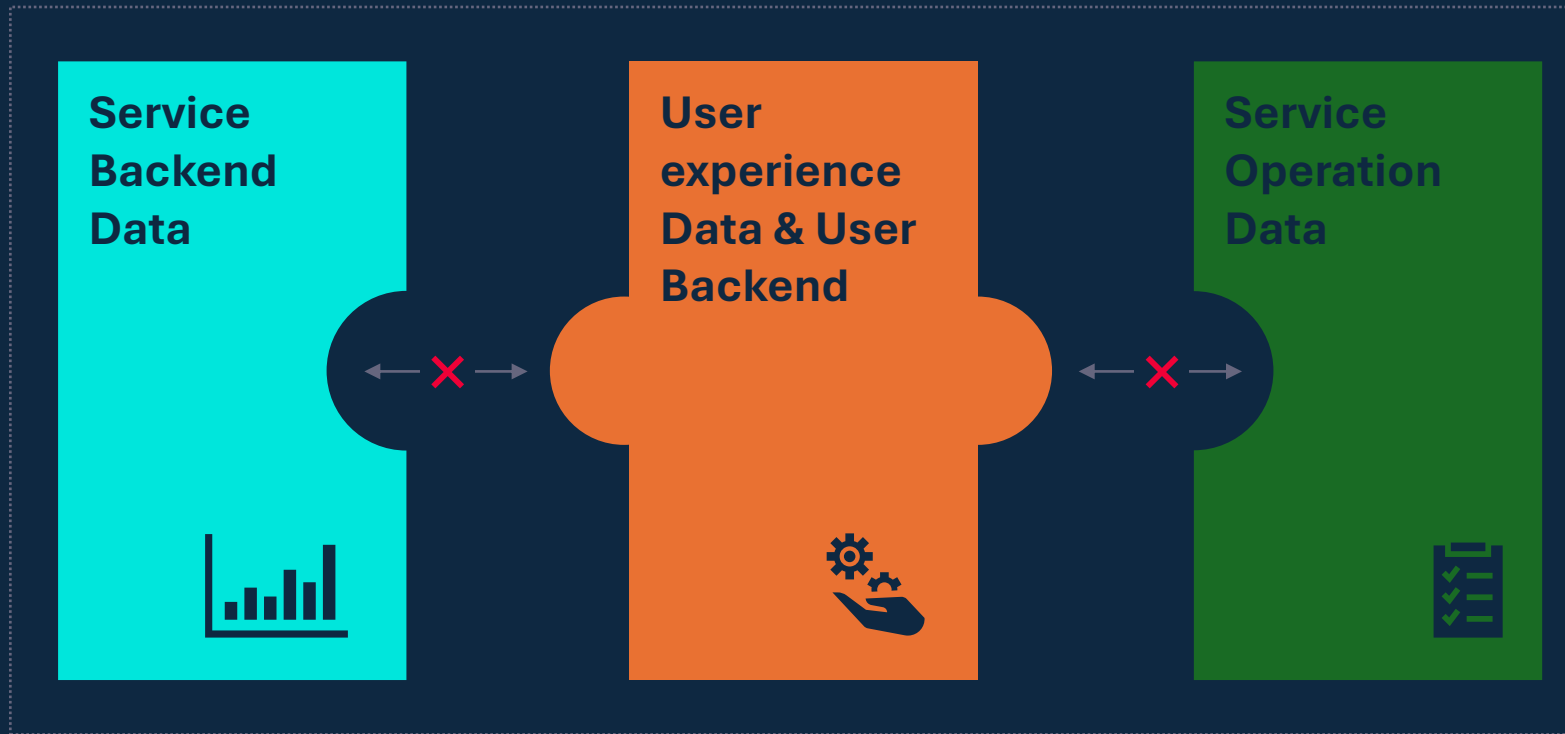
Cable temperature issues



Agenda

	Topic
1	Topic overview
2	Solution setup overview
3	Dataset creation and Log retrieval
4	Model, feature & parameter overview
5	Model training and results
6	Result verification
7	Lookahead and next steps
8	Links & References
9	Visualization and potential additional outlet filtering
10	Full_workflow script manual

Challenge today



From this follows:



Reactive not predictive

Data Silos without autom. Exchange

Painful processes

The magic happens when Copper + Code + Competence service work together



Cable Plugs – Wear and Tear

Cable overheating

- Cables are overheating and charging sessions interrupted due to this
- Temperature difference between the sensors (PT1 & PT2) differs increasingly
- Welding effect of plug and car as a worst case
- Intact cables are replaced out of fear
- Increased outlet downtime

Early detection

- Missing transparency into cable status
- Early notification on affected cables needed
- PLE Error flags only when the temperature difference between the two sensors reaches 25 degrees

Additional notifications

- Scalability to implement the device model with better and more detailed data
- Identification of outlier events (e.g. Pins get damaged due to strain on the cable)
- (Near) real time analysis for early reaction

Further Documentation and more detailed slides on this can be found on Confluence: Troubleshooting – SiD ([klick here to go to the detail page](#))

General setup

API Based Log Retrieval

- Using the TVM keys to access Cloudwatch in the scripts
- Select Query and Parameters
- Segmentation to cheaply create datasets
- Download as a Parquet file

Model training and evaluation

- Selection of three different models to evaluate results
- Train on the big, full year dataset
- Train on temperature diff prediction & identify spikes along the way

Data Preprocessing and feature engineering

- Creation of rolling features for the timeseries models
- Selection of parameters for the models
- Diff and Diff / Energy as parameters

Model Usage and problem identification

- Ingest newest data
- Create overviews of problem predictions for a specific time window
- Update the data regularly for newest alerts

Script overview

Log retrieval
(full year, all chargers
and new logs)

Log
combination
and cutoff

Creation of param.,
model, features and
metadata

Analysis on
new data

Comparison of
the results

Log Download

LSTM

LSTM_Prod

Log
Combination

RNN

RNN_Prod

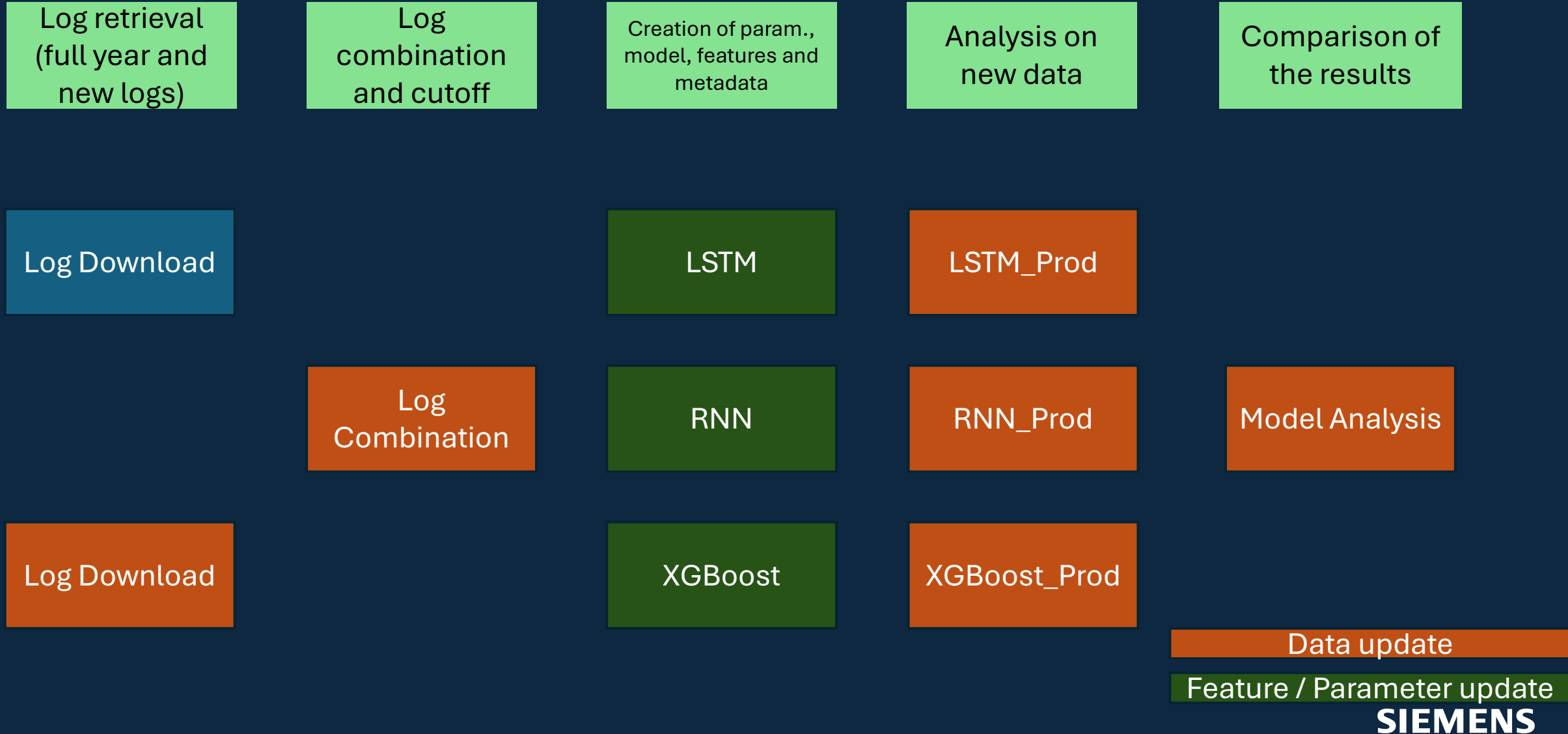
Model Analysis

Log Download

XGBoost

XGBoost_Prod

Updating the setup (Before model selection)



Cable Temperature tracking Logs

SeccSessionStop

- Providing temperature data along with other measurements after a finished charging session
- Only once value per session
- Logged in standard mode

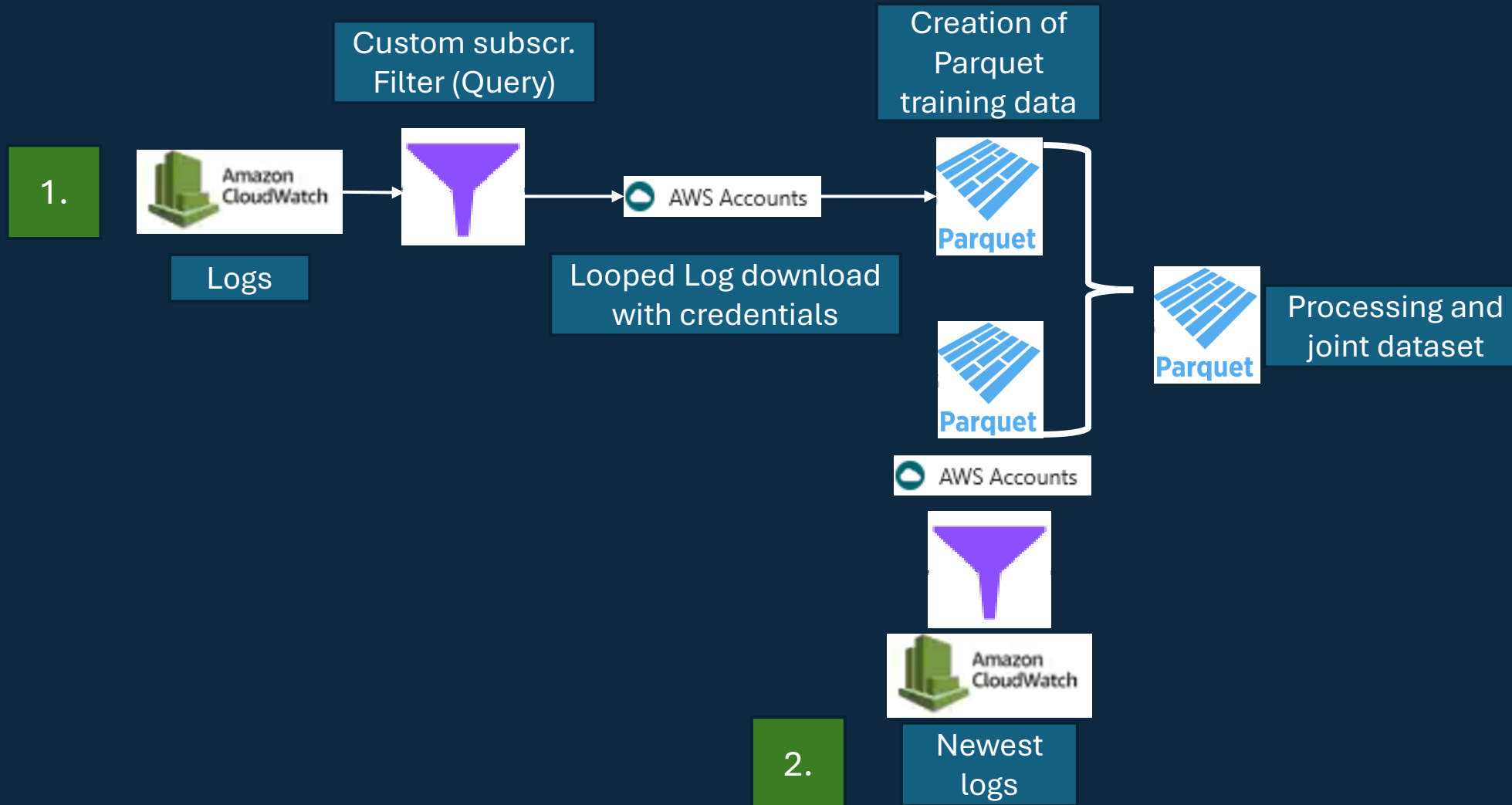
PLE & CTD

- Cable temperature sensors variance Flags
- Logged once when certain thresholds are exceeded (CTD for 5-, 10- & 15-degree variation between PT1 & PT2)
- PLE: 25-degree variation and charging session stopped

CLG

- Continuous sensor data logging throughout a charging session
- Data similar to the one attainable in the device model
- Active in Debug mode only

Current Setup for Log retrieval using credentials from the token vending machine



AWS Query

```
fields @timestamp, @message, @LogStream, @Log

#| filter @logStream like /0BTBMJ/
| filter @message like /SeccSessionStop/ or @message like /PLE:1/

| parse @message '"curr_ses_wh":*, ' as energy
| parse @message '"pt1000_1_T":*, ' as pt1000_1_T
fields @timestamp, @message, @LogStream, @Log

#| filter @logStream like /0BTBMJ/
| filter @message like /SeccSessionStop/ or @message like /PLE:1/

| parse @message '"curr_ses_wh":*, ' as energy
| parse @message '"pt1000_1_T":*, ' as pt1000_1_T
| parse @message '"pt1000_2_T":*, ' as pt1000_2_T
| parse @message '"outlet":*, ' as outlet1
| parse @message '"evsestat":*, ' as evsestat
| parse @message '"curr_ses_secs":*, ' as duration
| parse @message /\\"EVMAC\\":\\"(?<EVMAC>.*?)\\"/
| parse @message "PLC-Error-Flag: '*_CableTempSensor_XD100'*" as extracted_id1
| parse @message "PLC-Error-Flag: '*_CableTempSensor_XD101'*" as extracted_id2
| parse @message "PLC-Error-Flag: '*_CableTempSensor_XD230'*" as extracted_id3
| parse @message "PLC-Error-Flag: '*_CableTempSensor_XD240'*" as extracted_id4
| parse @message "PLC-* -Flag" as Error
| parse @message '"initializing":*, ' as False

| parse extracted_id1 /(?:[A-Z]+)?(?<extracted_number1>\d+)/
| parse extracted_id2 /(?:[A-Z]+)?(?<extracted_number2>\d+)/
| parse extracted_id3 /(?:[A-Z]+)?(?<extracted_number3>\d+)/
| parse extracted_id4 /(?:[A-Z]+)?(?<extracted_number4>\d+)/
| parse outlet1 /(?:[A-Z]+)?(?<outlet>\d+)/

| display concat (@LogStream, outlet, extracted_number1, extracted_number2,
extracted_number3, extracted_number4) as IDoutlet, @timestamp, abs(pt1000_2_T-pt1000_1_T) as diff,
((diff*1000)/energy) as Energydiff, concat(Error, False) as PLE_Error

| sort by @timestamp desc
| limit 10000
```

Filter for Energy transferred and minimum 1-minute charging sessions

Filter for PLE Errors and Charging Data

Definition of Parameters for the table format:

- energy
- PT1 & PT2
- Duration
- EV Address and EV Stat
- Outlet (for both messages)
- PLE Error indicator

Creation of table, custom values and combining values for training data

AWS Query results

SeccSessionStop						
#	IDOutlet	1 @timestamp	2 diff	3 Energydiff	PLE_Error	4
▶ 1	BjXnH11	2025-08-04T10:35:59.240Z	1.5548	0.06524	false	
▶ 2	Nz1TKf1	2025-08-04T10:35:46.057Z	0.7774	0.05302	false	
▶ 1	jdQYr82	2025-08-04T10:36:06.167Z			Error	PLE Error

1 Outlet + ID (combined) | Timestamp

2 Diff: Temperature difference of PT1 and PT2 after the charging session

3 Energy transferred is considered:
Energydiff: |PT1-PT2| / Energy
(diff*1000) / energy

4 PLE Error indication

From Mid August 2024 to Mid August 2025:
4.944.745 logs (90% SeccSessionStop)

Total of 32.219 daily logs (04.08.25): 87% SeccSessionStop
Total of 880.658 monthly logs (July 25): 92% SessSessionStop

Model overview

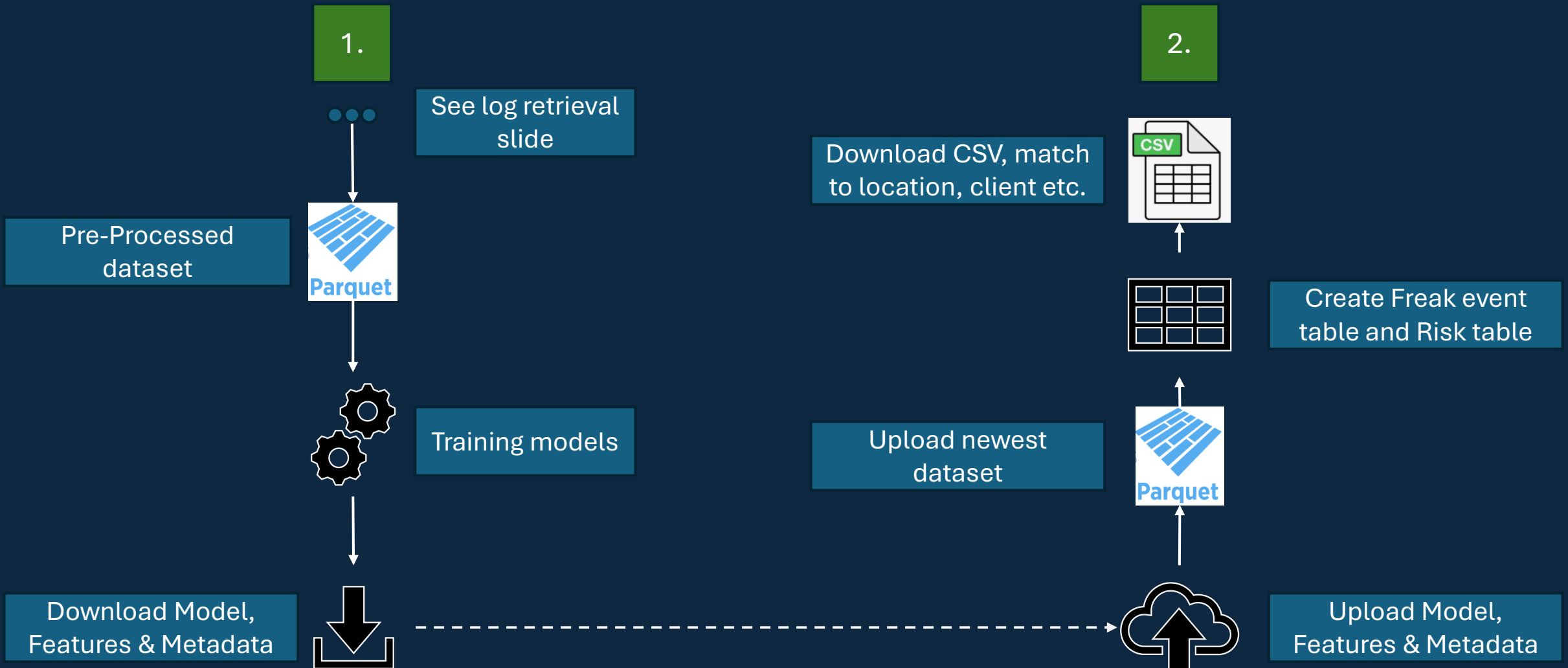
Spec	Keras LSTM	Keras RNN	XGBoost
Sustainability for incomplete data	Low – relies on full, homogeneous sequ.	Medium – can handle short bursts with masking	High – single-point features only
Training complexity	High	Medium	Low
Interpretability	Low	Medium	High
Input flexibility	Low – needs fixed-length	Medium – variable-length with masing	High – no sequence constraints
Missing data handling	Poor – requires preprocessing	Medium – masking helps but gaps still tricky	Excellent – native missing-value support
Inference latency	Medium – sequential cell overhead	Low – simpler cell than LSTM	Very low – fast tree ensemble inference

Also providing better results after training

Models

RNN	XGBoost	LSTM
Keras	XGBoost	Keras
SimpleRNN(32, <i>return_sequences=True</i> , <i>dropout=0.3</i>)	objective = 'binary:logistic'	LSTM(64, <i>dropout=0.2</i>)
SimpleRNN(16, <i>dropout=0.3</i>)	max_depth = 6	Dense(32, <i>activation='relu'</i> , <i>dropout=0.2</i>)
Dense(16, <i>dropout=0.2</i>)	min_child_weight = 3	Dense(1, <i>activation='sigmoid'</i>)
Dense(1, <i>activation='sigmoid'</i>)	subsample = 0.8	# Memory Optimization MAX_SEQUENCES_ASSET = 200
batch_size = 64	colsample_bytree = 0.8	MAX_TOTAL_SEQUENCES = 100000
epochs = 50	learning_rate = 0.1	BATCH_SIZE = 32
optimizer = 'adam'	gamma = 0.1	SEQUENCE_SAMPLING_RATE = 0.7
loss = 'binary_crossentropy'	n_estimators = 200	MEMORY_EFFICIENT_MODE = True
early_stopping_patience = 10	early_stopping_rounds = 20	TIMESTEPS = 20

Training and Analysis



Features

Parameter	Description
diff	Temperature difference of PT1 & PT2 after the charging session
Spike_flag	Sudden temperature change (e.g. pin gets damaged at charging session)
Reset_flag	Reset (e.g. cable change) detected
sessions_since_reset	Sessions since reset (e.g. cable change)
Energydiff	Difference / Energy per session (to factor in duration / energy for each session)
Warning labels	Mild, Medium & Critical Error Flags
LSTM add. Features	Rolling mean over 7 days, rolling standard dev., rolling min and mx

Labels and Parameters for the selected RNN model

Parameter	Current Value	Description
Rolling_window	150	Number of sessions to look back for rolling calculations and failure pattern identification
DIFF_Threshold	12.0	Temperature difference threshold for failure detection
Perc._Threshold	0.2	Percentage of Sessions threshold to trigger label
Spike_Threshold	4.0	Temperature spike detection threshold (middle of rolling window)
Reset_Threshold	80%	Drop threshold for detecting resets (e.g. cable changed or cleaned)
Min_sessions_ before_reset	10	Minimum sessions before considering a reset
Tier warning system	4, 8, 12	Tier based warning system, more sensitive to severity

Two options for implementing predictive Maintenance were investigated

“Fake” Forecasting

- Identification of feature-based patterns in rolling windows with thresholds low enough to ensure time lag to real failure
- Theoretically prone to be solved with rule-based models (but very poor performance on this dataset)
- Timestamp in the past x days, where a pattern of early temperature difference (together with patterns in the other features) is identified
- Presumably, shorter time to respond before the failure occurs

Real forecasting

- Forecasting based on pattern identification of the features to specific thresholds
- Earlier issue identification
- Lower model performance due to unclean dataset and no “proper” timeseries data
- Prone to be adapted for multiple features with the implementation of the device model and continuous sensor data
- Warning of a future breach of thresholds

Model performance overview for “fake” forecasting

Model	Rolling Window Sessions	diff threshold	percentage threshold	Spike Threshold	Performance	test_acc	test_pre	test_recall	test_f1
XGBoost	60	10	0.3	5	Test dataset	0.99	0.51	0.98	0.67
XGBoost	90	8	0.5	5	Test dataset	0.998	0.62	0.88	0.73
XGBoost	150	12	0.4	4	Test dataset	0.996	0.38	0.77	0.51
XGBoost	150	12	0.2	4	Test dataset	0.99	0.56	0.88	0.69
RNN	30	6	0.25	4	Test dataset	0.98	0.61	0.92	0.74
RNN	120	7	0.35	3.5	Test dataset	0.99	0.35	0.83	0.49
RNN	105	7	0.2	4	Test dataset	0.95	0.25	0.69	0.37
RNN	150	12	0.2	4	Test dataset	1	0.85	0.87	0.86
RNN	150	15	0.2	4	Test dataset	1	0.68	0.87	0.77
LSTM	20	5	0.2	2	Test dataset	0.98	0.96	0.74	0.83

*The tried-out rule-based model did not identify outlets sufficiently well and was disregarded

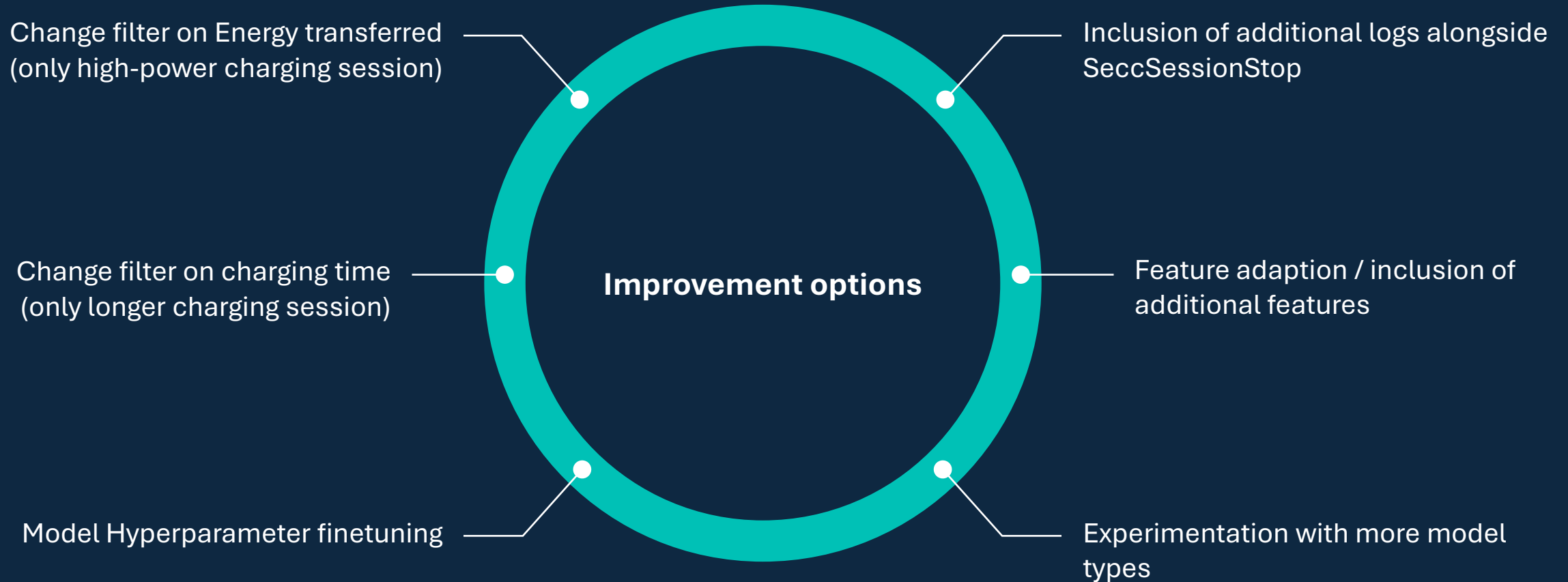
Model performance overview for “real” forecasting

Model	Rolling Window Sessions	Prediction horizon days	diff threshold	percentage threshold	Spike Threshold	Performance	test_acc	test_pre	test_recall	test_f1
XGBoost	120	90	8	0.2	2	Test dataset	0.85	0.61	0.55	0.58
RNN*	120	30	7	0.2	4	Test dataset	0.32	0.66	0.5	0.25

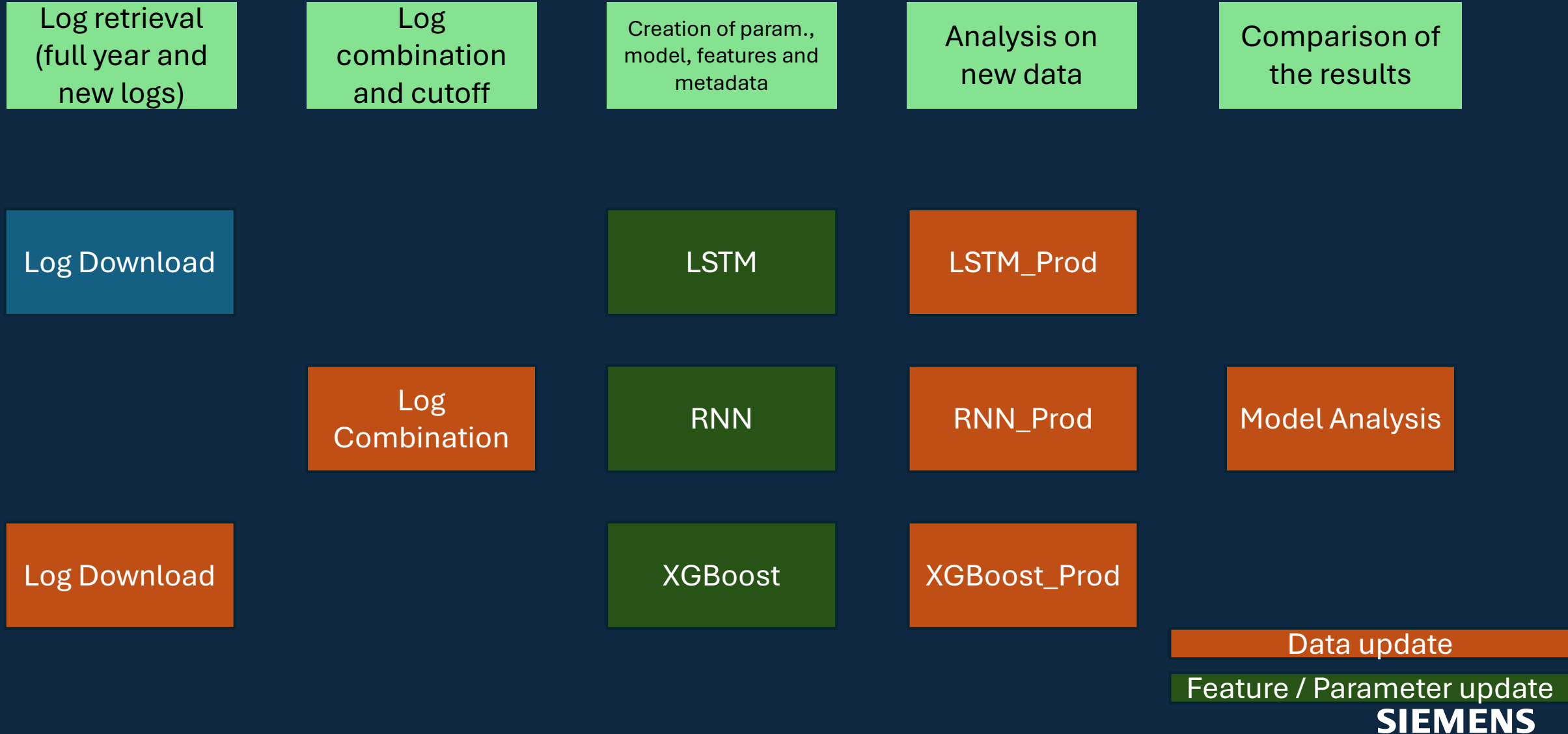
A lot more training could be carried out for hyperparameter finetuning and parameter optimization (scripts available in OneDrive)

*Multiple other variations of RNN configurations showed similar results -> Focus on XGBoost

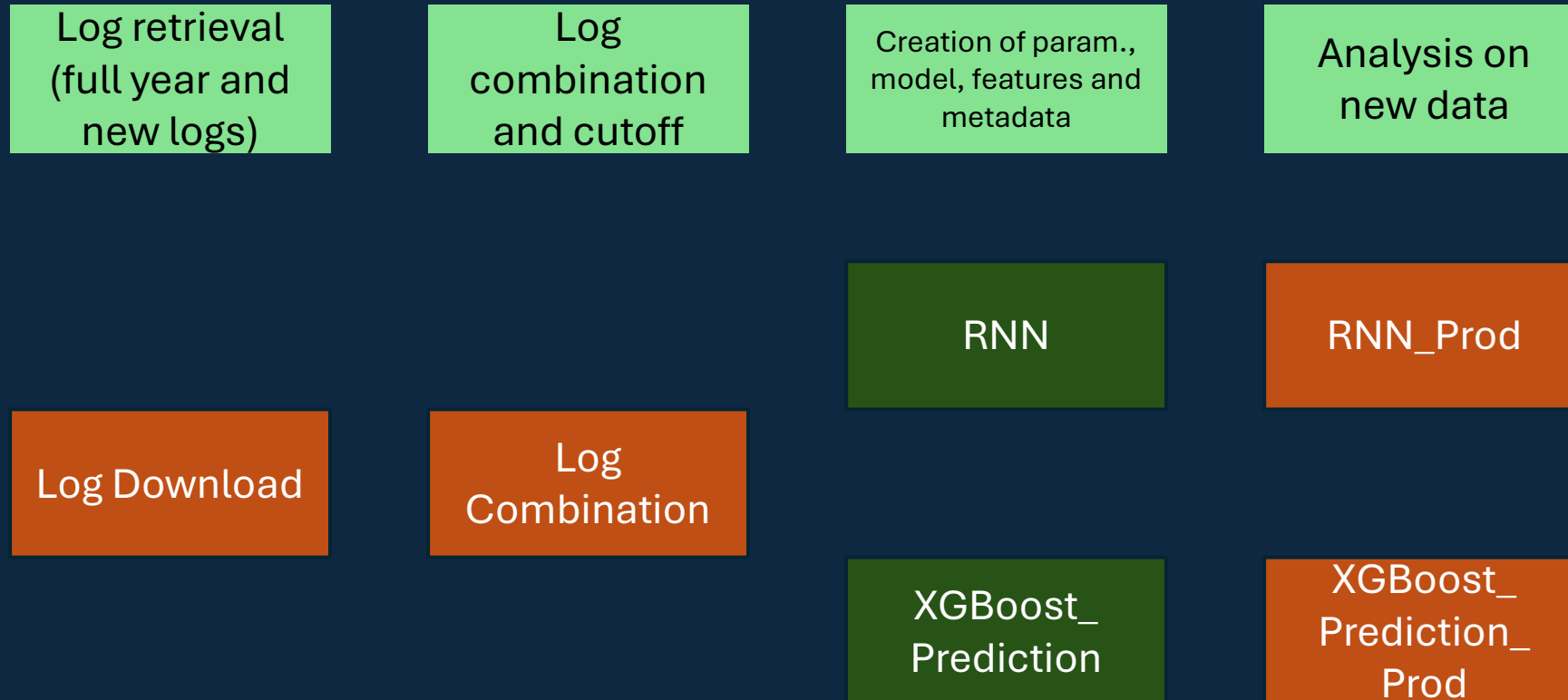
Ideas for further performance improvement



Updating the setup (Before model selection)



Updating the setup (After model selection)



Data update

Feature / Parameter update

SIEMENS

Simplified Setup: All in one

Log retrieval
(full year and
new logs)

Log
combination
and cutoff

Creation of param.,
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Analysis on
new data

Log Download

Log
Combination

RNN

RNN_Prod

XGBoost_
Prediction

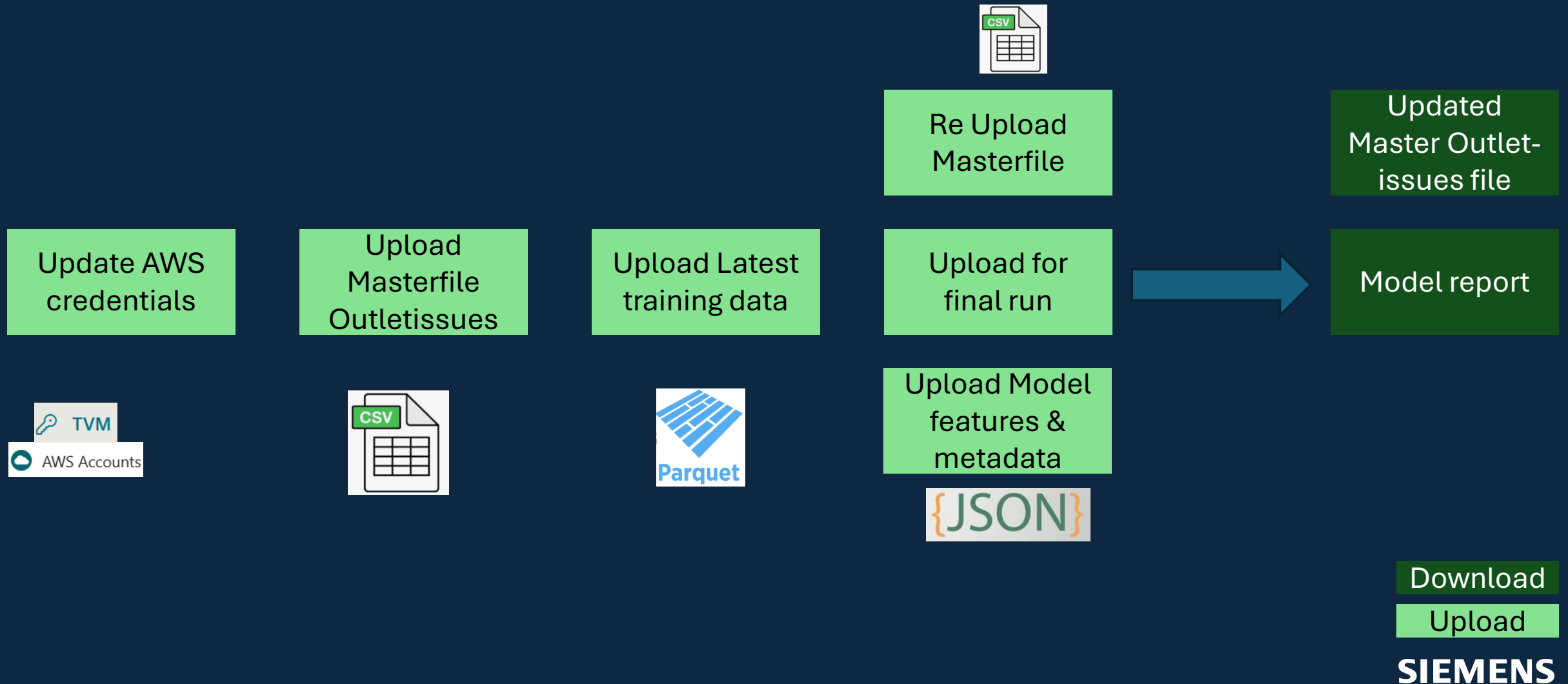
XGBoost_
Prediction_
Prod

Data update

Feature / Parameter update

SIEMENS

Simplified Setup: Reporting all in one script (training is extra)



Results / Output for 30 days looking back in February

Summary

Analysis Period	Last 30 Days
Analysis Date	2025-02-26 15:33:15
DATA OVERVIEW	
Total Records Analyzed	802355
Unique Outlets Analyzed	4300
FREAK EVENTS	
Total Freak Events Detected	87
Critical Freak Events (≥95%)	79
High Freak Events (90-94%)	4
Medium Freak Events (80-89%)	4
RISK CATEGORIES	
Total Risk Predictions	432
Critical Risk (≥90%)	84
High Risk (80-89%)	9
Medium Risk (60-79%)	339
THRESHOLDS USED	
Freak Event Threshold	≥80%
Critical Risk Threshold	≥90%
High Risk Threshold	≥80%
Medium Risk Threshold	≥60%

Freak event and Risk table

Outlet ID	Freak Severity (%)	Detection Date	Risk Level	Prediction Probability
qbAXSF1	100	2025-02-24 09:44	CRITICAL	1
yRk5yL3	100	2025-02-24 11:53	CRITICAL	1
7UU7cO1	100	2025-02-24 13:18	CRITICAL	1
j8jAoJ3	100	2025-02-24 17:50	CRITICAL	1
Q4vKdH3	100	2025-02-24 17:33	CRITICAL	1
mAiNo42	100	2025-02-24 17:23	CRITICAL	1
YB7uye3	100	2025-02-24 16:35	CRITICAL	1
rn5pA91	100	2025-02-24 21:42	CRITICAL	1
DqQLib2	100	2025-02-24 19:53	CRITICAL	1

Category	Outlet ID	Prediction (%)	Detection Date	Action Required	Prediction Probability
Critical Risk	fMiFUM1	100	2025-02-24 23:56	IMMEDIATE	1
Critical Risk	rn5pA91	100	2025-02-24 21:42	IMMEDIATE	1
Critical Risk	uEltpP2	100	2025-02-24 21:34	IMMEDIATE	1
Critical Risk	ObibH92	100	2025-02-24 20:10	IMMEDIATE	1
Critical Risk	oLG0BP1	98.3	2025-02-24 20:04	IMMEDIATE	0.983
Critical Risk	DqQLib2	100	2025-02-24 19:53	IMMEDIATE	1
Critical Risk	vrOFT61	98.7	2025-02-24 19:39	IMMEDIATE	0.987
Critical Risk	x4Uawq2	100	2025-02-24 18:58	IMMEDIATE	1
Critical Risk	uoQAdw2	100	2025-02-24 18:55	IMMEDIATE	1
Medium Risk	WAuhak2	71.4	2025-02-24 18:32	MONITOR	0.714

For a continuous monitoring and reporting, a master file was also created to track past and new issues

Master List

Outlet ID	ID	Outlet	Detection Date	Counter
0AQbZI1	0AQbZI	1	2025-09-05 06:55	5
0PrWyp3	0PrWyp	3	2025-09-05 06:53	5
17xsOj2	17xsOj	2	2025-08-20 16:52	9
198r072	198r07	2	2025-09-05 01:21	13
198r074	198r07	4	2025-08-27 11:45	5
1h4b1k1	1h4b1k	1	2025-08-03 07:46	1
28JUhb1	28JUhb	1	2025-09-05 07:22	9
2bBEih1	2bBEih	1	2025-09-01 15:48	9
2cb4QG2	2cb4QG	2	2025-08-29 13:07	5
32CPmS1	32CPmS	1	2025-08-10 17:17	5
32Dbgv1	32Dbgv	1	2025-08-17 09:55	5
3g38Q64	3g38Q6	4	2025-08-11 10:39	5
3Ji96w2	3Ji96w	2	2025-09-03 16:09	9
3O4RJk2	3O4RJk	2	2025-08-13 21:02	5
4ChOAC2	4ChOAC	2	2025-08-31 07:34	5
4HlbKg1	4HlbKg	1	2025-08-08 05:20	5
50OR541	50OR54	1	2025-08-18 04:34	5
5R9akG2	5R9akG	2	2025-09-04 19:22	9
6GnOnr1	6GnOnr	1	2025-08-10 15:02	5
6GnOnr2	6GnOnr	2	2025-08-08 09:51	5

For further analysis the report selects:

- Top 10 counted outlets
- 50 newest outlets
- All newly identified outlets

Automized second selection based on KPIs for the identified outlets

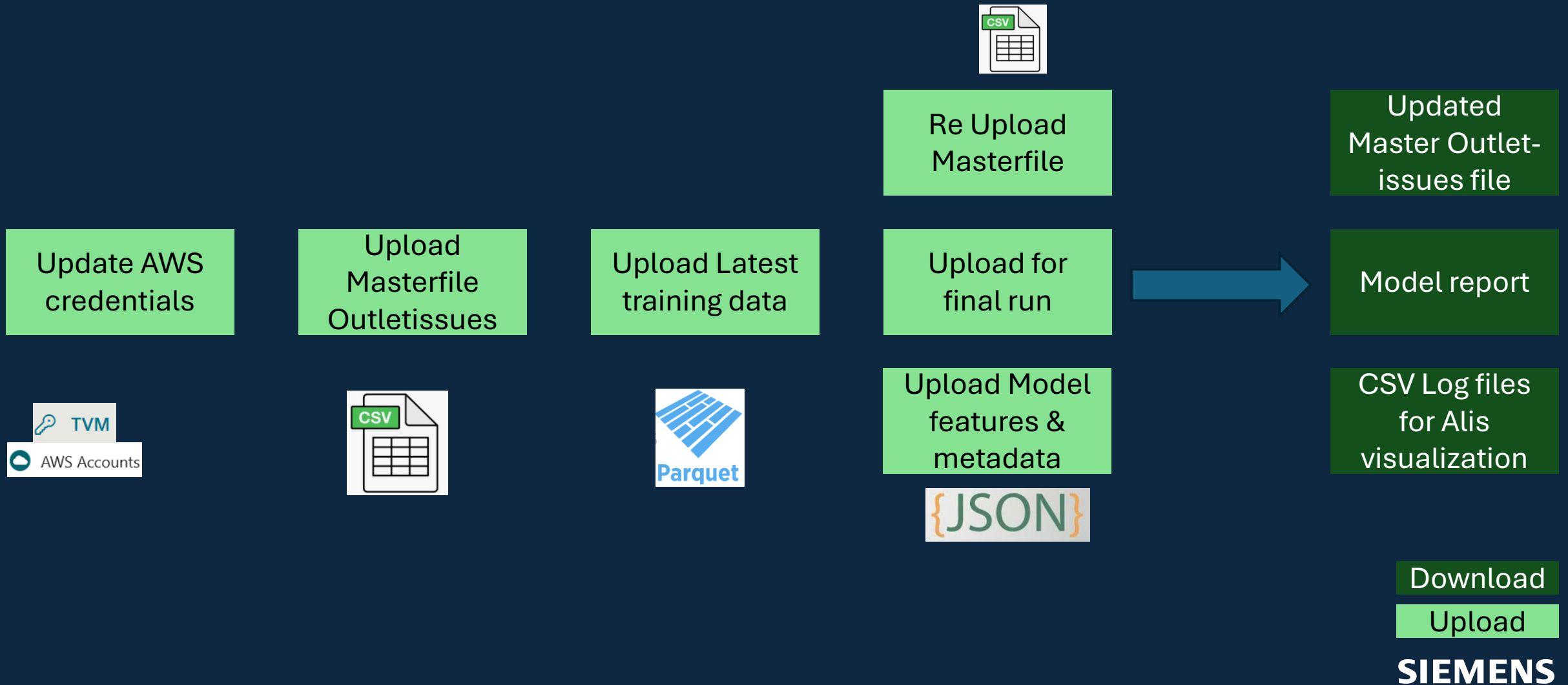
50 newest timestamps of identified issues

Outlet ID	ID	Outlet	Detection Date
uoQAdw2	uoQAdw	2	2025-09-09 13:31
DViUun1	DViUun	1	2025-09-09 12:58
28JUhb1	28JUhb	1	2025-09-09 12:38
ObibH92	ObibH9	2	2025-09-09 12:26
AT2CQ71	AT2CQ7	1	2025-09-09 12:22
XfClwM3	XfClwM	3	2025-09-09 12:16
sAUNV52	sAUNV5	2	2025-09-09 12:11
obYNIH1	obYNIH	1	2025-09-09 12:04
9KXfLP3	9KXfLP	3	2025-09-09 11:52

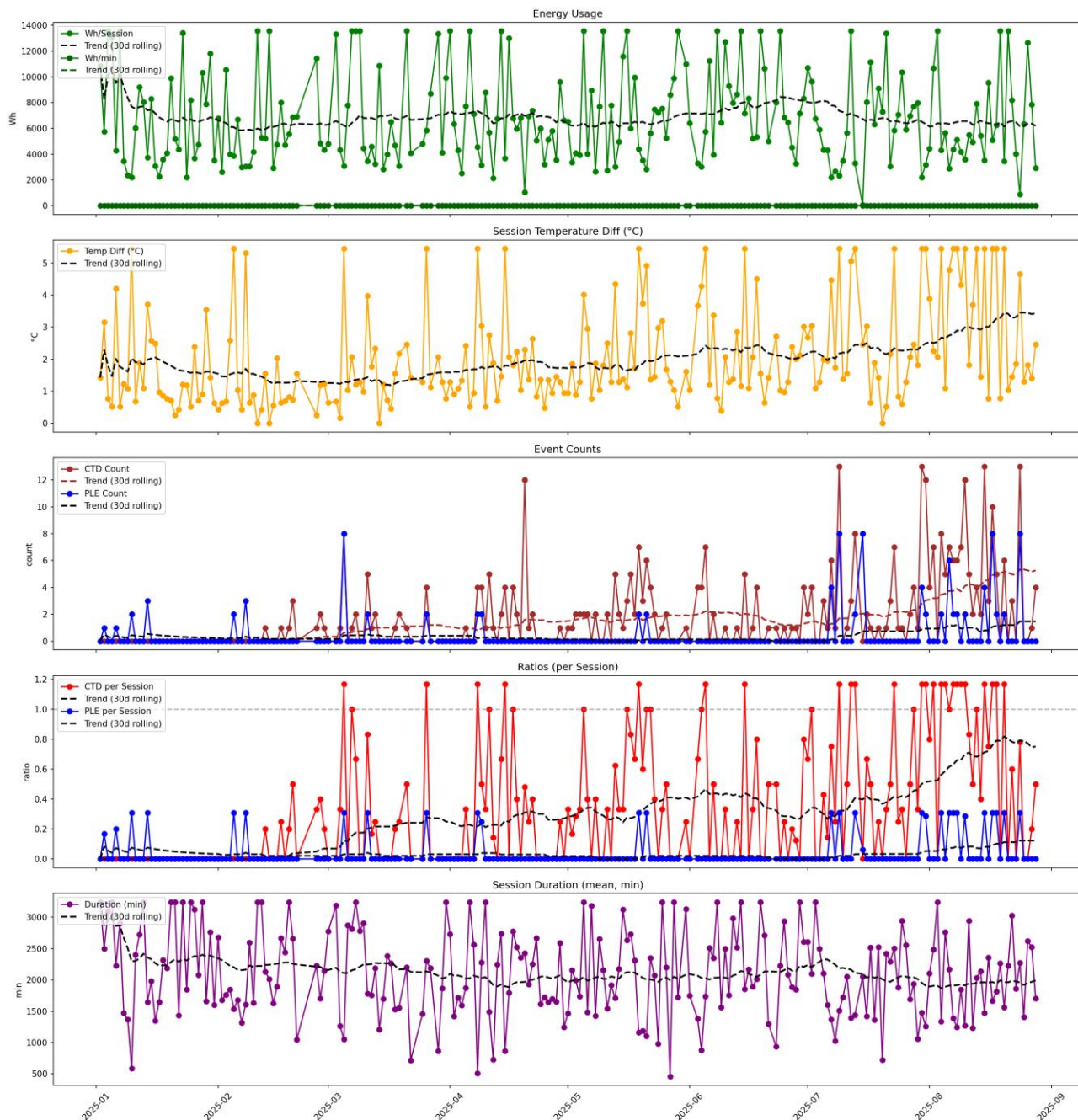
Newly identified issues

Outlet ID	ID	Outlet	Detection Date
AT2CQ73	AT2CQ7	3	2025-08-20 15:14
1m4Zhr1	1m4Zhr	1	2025-08-21 10:07
vnpSc21	vnpSc2	1	2025-08-24 15:23
hKzfTF4	hKzfTF	4	2025-08-25 15:19
1m4Zhr2	1m4Zhr	2	2025-08-26 16:22
U6H3Gj3	U6H3Gj	3	2025-09-01 07:19
Zt9lsW2	Zt9lsW	2	2025-09-01 13:53
ZacobH1	ZacobH	1	2025-09-02 16:43
YQf55z1	YQf55z	1	2025-09-03 12:22

Simplified Setup: Reporting all in one script (training is extra)



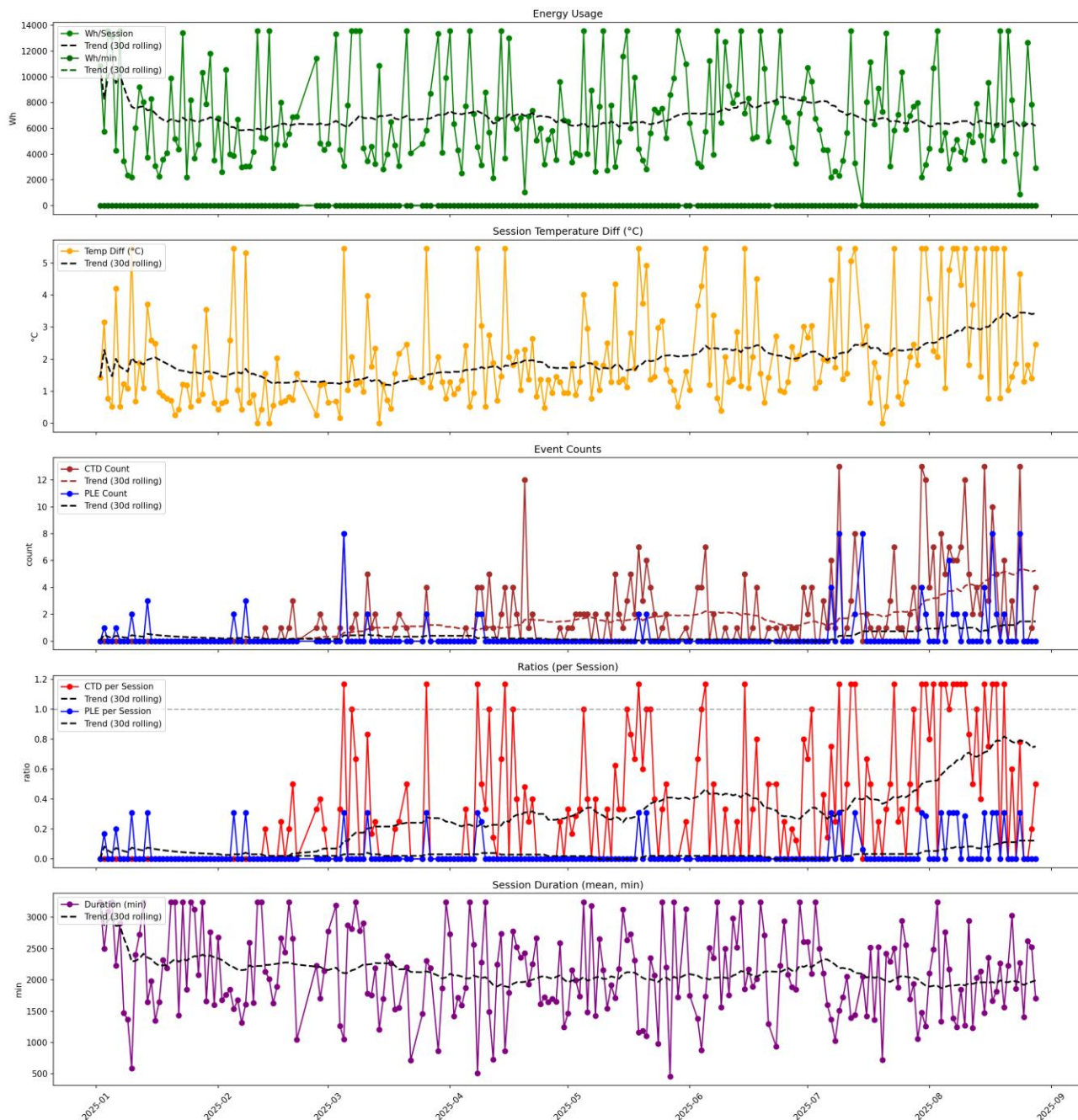
Second outlet selection is carried out with
Visualizations



Additional visualizations / selections with specific KPIs (in the report specific selection of most important outlets / newest outlets with issue):

- Energy Usage / Session
 - Temp Difference / Session
 - CTD and PLE count over time
 - CTD and PLE / Session
 - Duration / Session
-
- => Generally: A lot of the visuals support the issue identification statement of the model
 - => Goal: Identification when customer experience becomes impaired due to failing charger cables

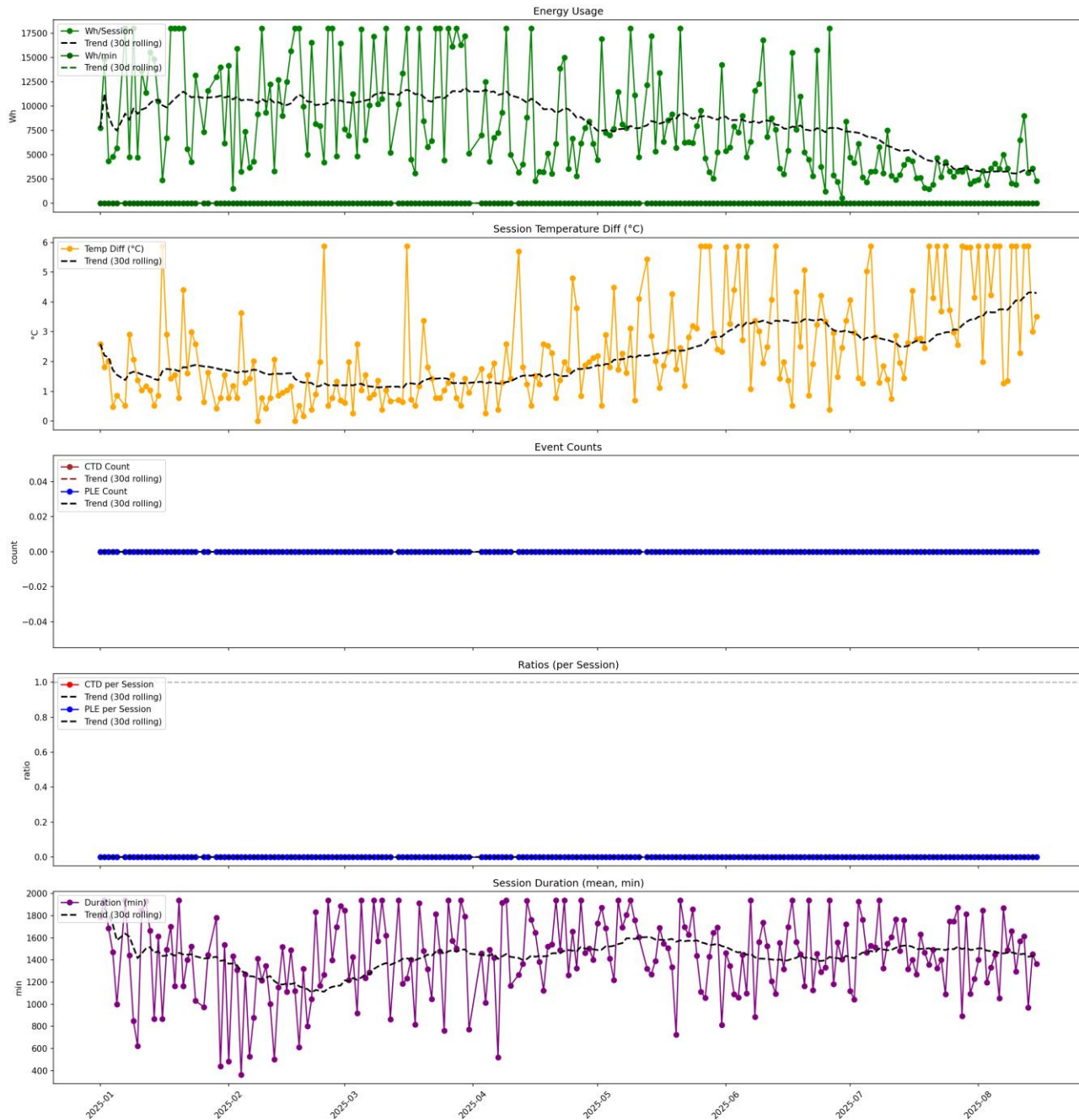
Timeline with Trends: 7jsgx4 - Outlet 1



=> Some of the outlets identified as faulty are showing a clear trend with higher urgency.

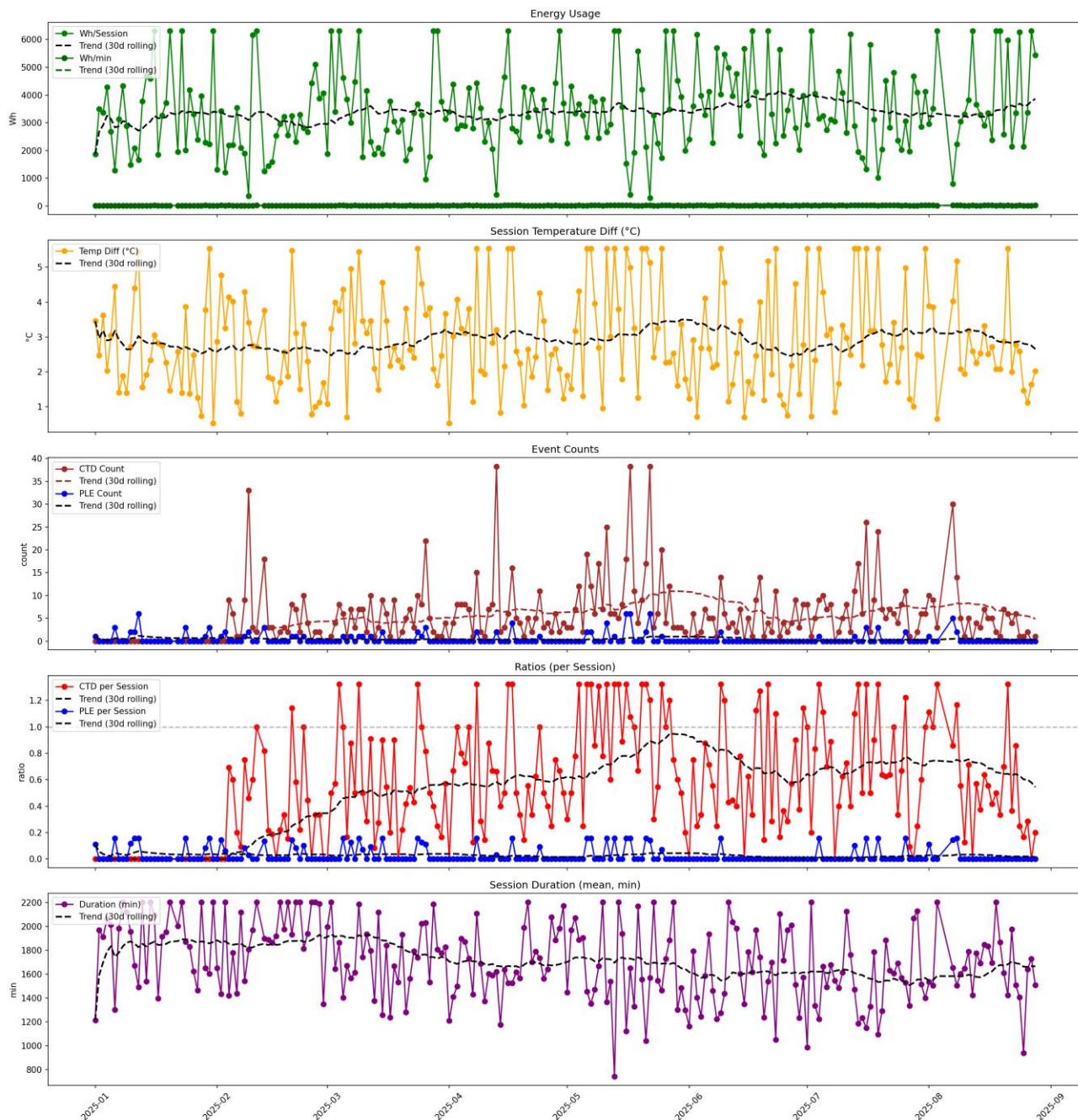
An increasing difference between the two sensors and number of CTD trigger warnings showcases that the outlet is trending towards failure. Still: Charging does not seem to be severely impaired (Energy and duration per session are fairly stable).

Timeline with Trends: sm6jea - Outlet 3



=> Some of the outlets identified as faulty are showing a clear trend with higher urgency

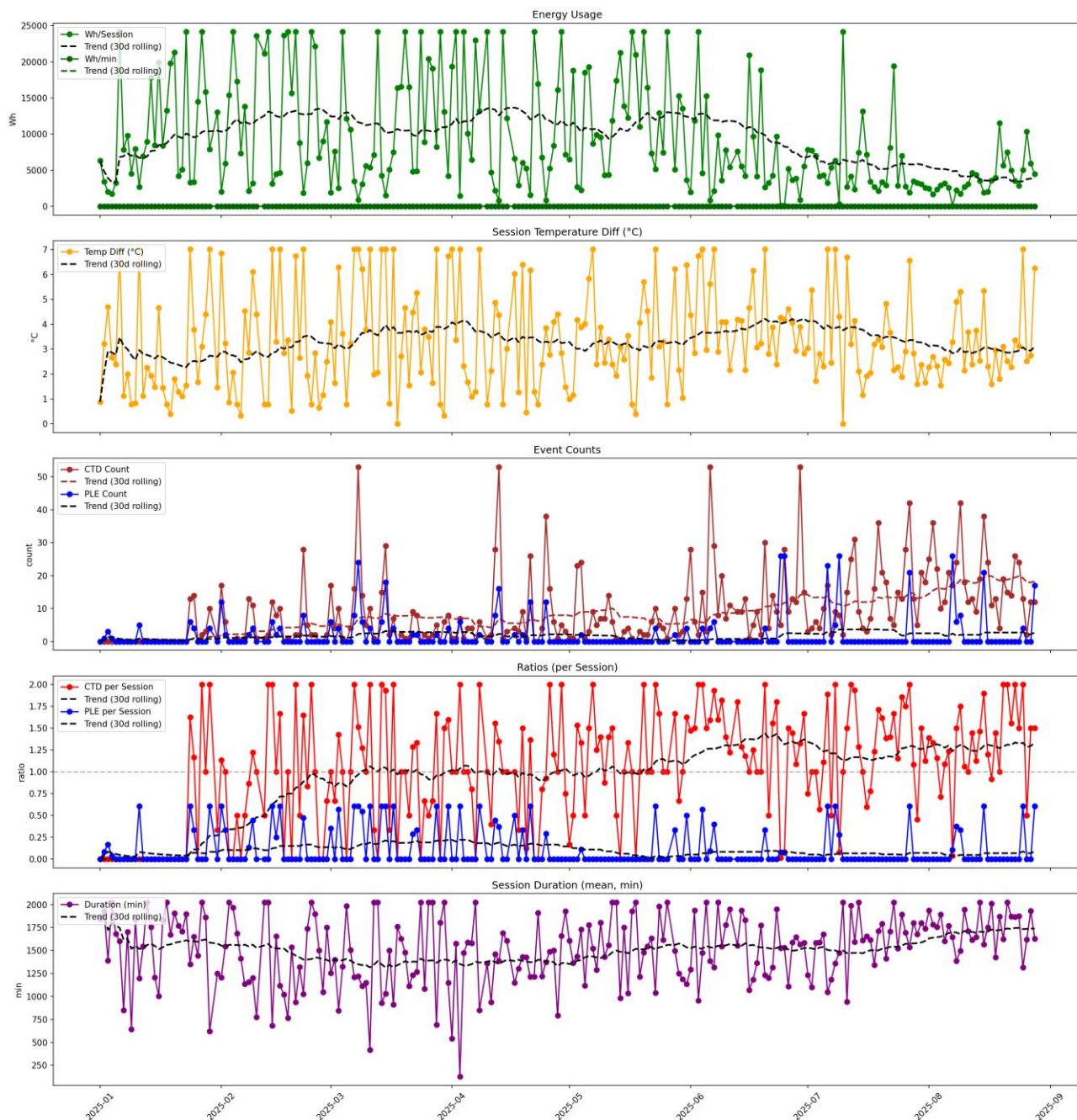
While CTD Logs are not filed for this outlet (presumably due to an older FW version), the trend for difference between the sensors and reduced energy indicate an increasingly bad charging experience and interruption.



=> Some of the outlets identified as faulty do not showcase any visual deterioration in the timeframe analyzed

The difference between the sensors is fairly high on average, also the amount of CTD warnings triggered. No impairment in terms of charging visible from those curves and not long-term deterioration trend visible either.

Timeline with Trends: 2cb4qg - Outlet 2



=> Some of the outlets identified are showing signs of issues for some of the metrics (deep dive recommended)

While the difference seems quite stable and only some more CTD flags are raised (although on a pretty high overall level), the energy goes down significantly. This could also indicate an unrelated other issue, and a deep dive would be recommended.

Issue verification rules to get the top 20 outlet issues based on these KPIs

CTD Score

- +2 points: $\max(\text{CTD_count}) > 50$ (large spikes)
- +1 point: $\text{median}(\text{CTD_count}) > 10$ (sustained smaller activity)
- Rationale: High/consistent CTD counts often precede cable failures.

Temperature Score

- +1 point: $\text{median}(\text{Sess_temp_diff_mean}) > 5^{\circ}\text{C}$
- +2 points: Sustained median $> 10^{\circ}\text{C}$ for 2–3 months
- Rationale: Failed cables often showed rising temp difference trends.

PLE Score

- +1 point: PLE_count shows multiple bursts > 5 (sporadic)
- +2 points: PLE_count shows bursts > 20 (serious anomaly)
- Rationale: PLE spikes indicate sensor/thermal plausibility issues linked to overheating.

Pattern Score

- +2 points: Temp diff shows steady rising trend (~2–4 months)
- +1 point: Session duration shows gradual downward trend
- +1 point: CTD and/or PLE also rising in the same period
- Rationale: Mirrors behavior observed in real failed cables.

Validity Check, Session Count Filter:

- Exclude outlets with total $\text{Sess_count} < 600$ (approx. below 25th percentile).
- Rationale: Insufficient sessions lead to unreliable or random trend observations.

Weighting & Priority

- Each rule contributes to a final score (maximum ~10 points).
- Higher scores indicate higher priority for visualization and service team review.

Model verification with Engie Chargers

Engie Charger Overview 01.01.2025-01.06.2025:

- Engie's fleet of chargers
- Error likelihood rate computation:
$$\text{PLE Error count} / \text{successful sessions count}$$
- Charging failure likelihood as a cable failure indicator
- Most cables replacements end of April until mid May
- Testing hypothesis of early identification of issues

=> Checking the output when cutting the model's dataset end of January and comparison to a simple rule-based approach using just Diff as a feature

no	charger ID	DC outlet	Outlet ID	cable replaced at	Count of PLE Errors	Count of started sessions	Error Likelihood per Session	Average sessions / day	Latest PLE Error
1	8iwoBh	2	8iwoBh2		1448	912	158.77%	6.04	2025-04-
17	XnvqOe	2	XnvqOe	2025-04-28	145	341	42.52%	2.26	2025-04-
3	mgibgC	2	mgibgC	2025-04-30	379	928	40.84%	6.15	2025-05-
11	OFNgi3	2	OFNgi3	2025-04-24	175	438	39.95%	2.90	2025-04-
2	DqQLib	2	DqQLib	2025-05-05	433	1156	37.46%	7.66	2025-05-
12	9NAmV1	1	9NAmV11		165	512	32.23%	3.39	2025-03-
7	aJbfMb	1	aJbfMb	2025-05-06	196	663	29.56%	4.39	2025-04-
16	RFhsW1	1	RFhsW11		148	550	26.91%	3.64	2025-05-
9	zhSxBz	2	zhSxBz	2025-04-28	191	849	22.50%	5.62	2025-05-
5	D1HnIC	2	D1HnIC	2025-05-14	239	1119	21.36%	7.41	2025-05-
6	dNKI21	2	dNKI21 ???		221	1039	21.27%	6.88	2025-03-
4	Q2MW2c	2	Q2MW: ???		290	1387	20.91%	9.19	2025-03-
15	NKqOgs	2	NKqOgs2		156	876	17.81%	5.80	2025-05-
14	h37RLm	2	h37RLr	2025-04-24	159	936	16.99%	6.20	2025-04-
20	wl7KBo	1	wl7KBo1		111	662	16.77%	4.38	2025-05-
27	srvCLH	2	srvCLH2		77	463	16.63%	3.07	2025-05-
13	rn5pA9	1	rn5pA9	2025-04-30	164	1044	15.71%	6.91	2025-04-
18	TWJsED	2	TWJsED2		135	952	14.18%	6.30	2025-06-
19	2cb4QG	2	2cb4QG2		118	836	14.11%	5.54	2025-05-
23	OEa0a	2	OEa0a	2025-04-30	97	737	13.16%	4.88	2025-04-
24	P5AJKi	1	P5AJKi1		97	741	13.09%	4.91	2025-04-
8	BWF50L	2	BWF50	2025-05-06	195	1540	12.66%	10.20	2025-04-
55	M8i3IC	2	M8i3IC2		38	309	12.30%	2.05	2025-04-
71	GBIAhX	1	GBIAhX1		31	279	11.11%	1.85	2025-04-
26	h37RLm	1	h37RLr not replaced		83	757	10.96%	5.01	2025-05-
49	FgLax3	2	FgLax32		43	405	10.62%	2.68	2025-04-
10	9k3hNN	2	9k3hNN2		181	1766	10.25%	11.70	2025-04-
28	bw2qCk	1	bw2qC	2025-04-24	76	762	9.97%	5.05	2025-04-
30	mtdGiB	1	mtdGiB	2025-02-07	63	641	9.83%	4.25	2025-02-
22	rn5pA9	2	rn5pA9	2025-04-30	97	1007	9.63%	6.67	2025-04-

Metrics and verification based on Engie data

Predicted outlet failures by the model with data up to 25th of January



25.01.25

Result verification by calculating the average failure rate of Engie chargers flagged in late January vs. chargers not flagged

“What is the average failure rate in early July for outlets flagged by the model vs. outlets not flagged”

Σ Failure rate of flagged chargers

Amount of Engie chargers flagged

Σ Failure rate of non-flagged chargers

Amount of Engie chargers not flagged

Calculation of metrics and failed outlets in early June – Proof of failing outlets

Engie analysis

01.06.25

SIEMENS

Comparison: RNN vs. Rule based approach with data until late January

Rule based model might be flawed

	RNN	Rule-based approach
Average charging error likelihood of identified cable problems	13,45%	4,43%
Average charging error likelihood of outlets deemed OK	2,51%	4,21%

There seems to be a clear predictive quality to the RNN outlet failure identification, which cannot be observed by using the same parameters on a rule-based model

- In the 90-day window predicted failures until late Jan showed a failure rate of 13,45 percent by early June
- Outlets not identified only had a ration of 2,51 PLE Errors per charging sessions

Additional factors to consider:

- Outlets with less than 150 charging sessions are not considered by the model
- Earlier replacements might have led to one of the reset flags in the model
- Model “certainties” seem a bit unrelated / unusable
- cable change time != time of failure

Metrics and verification based on Engie data

Predicted outlet failures by the model with data up to 25th of January



25.01.25

Result verification by comparing the number of predicted failures to certain failure threshold (e.g. failure threshold 15% + in early June)

“What percentage of outlets, failing in 15% of all sessions or more by June was already flagged in late January”

no	charger ID	DC outlet	Outlet ID	cable replaced at	Count of PLE Errors	Count of started session	Error Likelihood per Session	A
1	8iwoBh	2	8iwoBh2		1448	912	158.77%	
17	XnvqOe	2	XnvqOe	2025-04-28	145	341	42.52%	
3	mgibgC	2	mgibgC	2025-04-30	379	928	40.84%	
11	OFNgi3	2	OFNgi3	2025-04-24	175	438	39.95%	
2	DqQLib	2	DqQLib	2025-05-05	433	1156	37.46%	
12	9NAmV1	1	9NAmV11		165	512	32.23%	
7	aJbfMb	1	aJbfMb	2025-05-06	196	663	29.56%	
16	RFhsW1	1	RFhsW11		148	550	26.91%	
9	zhSxBz	2	zhSxBz	2025-04-28	191	849	22.50%	
5	D1HnIC	2	D1HnIC	2025-05-14	239	1119	21.36%	
6	dNKI21	2	dNKI21 ???		221	1039	21.27%	
4	Q2MW2c	2	Q2MW: ???		290	1387	20.91%	
15	NKqOgs	2	NKqOgs2		156	876	17.81%	
14	h37RLm	2	h37RLr	2025-04-24	159	936	16.99%	
20	wl7KBo	1	wl7KBo1		111	662	16.77%	
27	srvCLH	2	srvCLH2		77	463	16.63%	
13	rn5pA9	1	rn5pA9	2025-04-30	164	1044	15.71%	
18	TWJsED	2	TWJsED2		135	952	14.18%	
19	2cb4QG	2	2cb4QG2		118	836	14.11%	
23	OEAO0a	2	OEAO0a	2025-04-30	97	737	13.16%	
24	P5AJKi	1	P5AJKi1		97	741	13.09%	
8	BWF50L	2	BWF50L	2025-05-06	195	1540	12.66%	
55	M8i3IC	2	M8i3IC2		38	309	12.30%	
71	GBIAhX	1	GBIAhX1		31	279	11.11%	
26	h37RLm	1	h37RLr not replaced		83	757	10.96%	
49	FgLax3	2	FgLax32		43	405	10.62%	

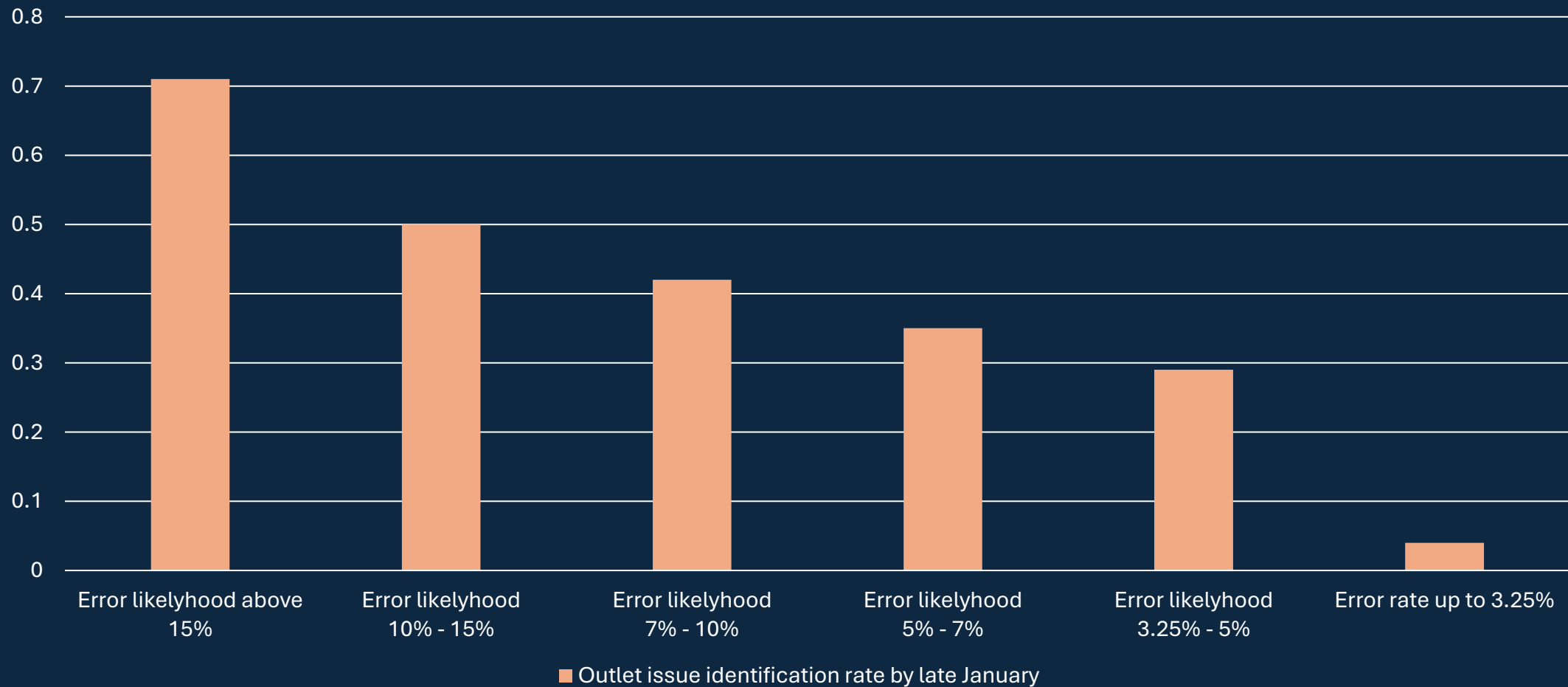
Calculation of metrics and failed outlets in early June – Proof of failing outlets

Engie analysis

01.06.25

SIEMENS

RNN Result verification with Engie Chargers on June 1st (Model predictions with Data until late January)



Conclusion

Current Assessment

- Model provides useful but not completely reliable output
- Possibility for pattern identification is there
- Useful support for early identification of outlet failures

Model optimization potential

- MLOps, proper finetuning
- Experimentation on true predictive model
- Potentially using a GPU to train a proper LSTM
- **Feedback loop for false positives (and potentially negatives)**

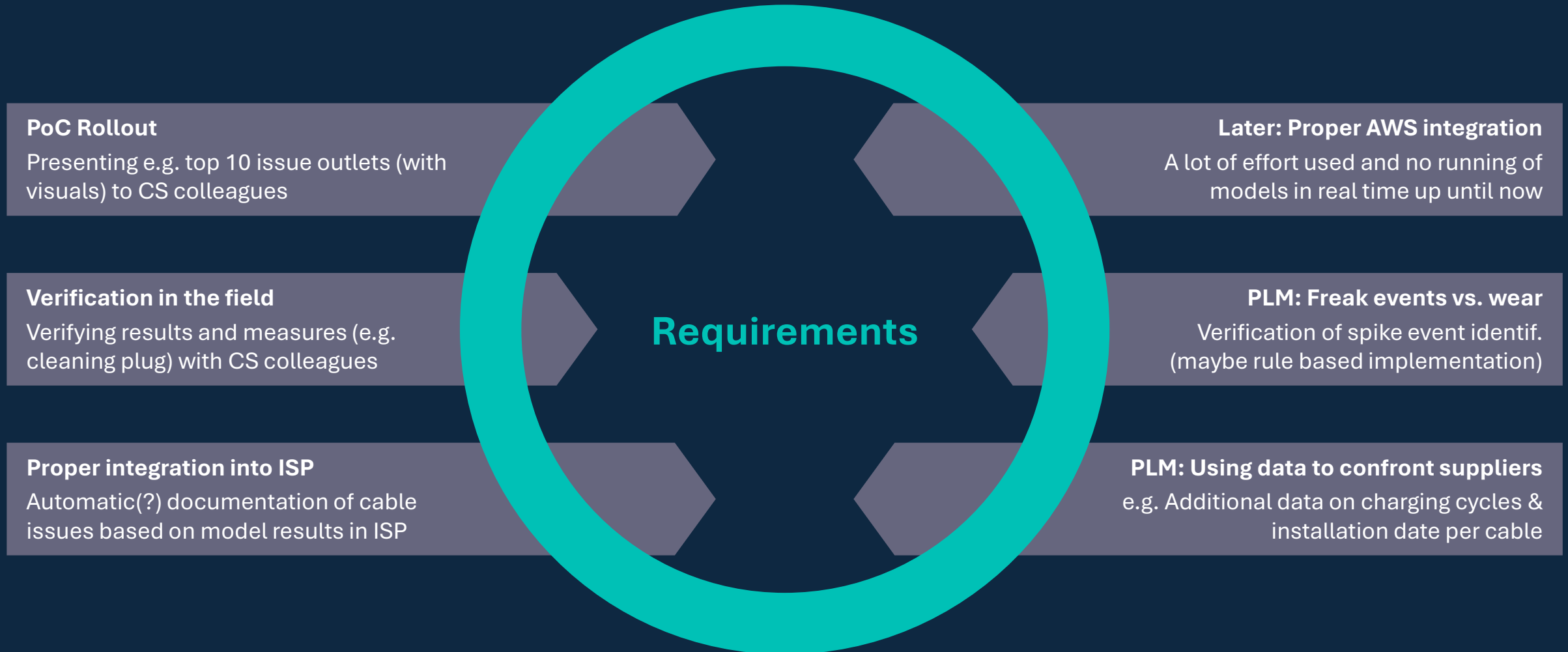
Key Takeaways

- Scaling possible but a bit limited
- A lot of preprocessing, AWS query creation and data transformation needed
- Actual AI part is rather small for now

Bottlenecks

- Time resources (full time almost three months for this workflow)
- Lack of AI engineering expertise
- Useful data availability issues

Additional functionality wishes and requirements



Device model needs and next steps

Data needs

- PT1 and PT2 (ideally logged throughout the charging session but at least at the end)
- Energy transferred in the session
- Duration of session
- Breaching of 5-, 10-, 15- and 25-degree sensor difference (if PT1 and PT only available after the session)
- Cross reference of those values (Device model values with charging session values)
- Ideally also outside temperature in relation and PT1 and PT2 at least at the start of the session

Next steps

- Selection of ten clearly compromised outlets
- Parallely: Second threshold-based filter on the KPI visualization
- Discussion with CS on usage and verification of top ten failing outlets in the field
- Experimentation with different techniques (cleaning outlets etc.)
- Adjust both model and visuals based on wishes and requirements of service engineers to ensure usability
- Roll out to more issues and use learnings from this project

Links & References

Jupyter Notebooks

**Cable overheating issue –
Confluence page**

Models

**Matching file (customer, country,
S/N based on SCB and SC data)**

Performance overview

Engie result verification file

Log files (Parquet)

Cloudwatch Query

Niklas/Predictive_Table

Log information Confluence Page

Onedrive structure

Scripts (Scripts for model training, Log retrieval & formatting and Analyzing Logfiles with pretrained models)



Cable Predictive Maintenance Proposal (Main Documentation and manual)



Models
(Performance overview and saved models from past runs with different parameters (assessment in Performance overview))



Logs
(SeccSession and PLE dataset for Logs from August 2024, cut off at different times)



Backup



Predictive Maintenance



Full Report Script and Docs
(Long script for the full report, newest Parquet file and newest Masterfile)



Output (Specific Logs for Alis model, Engie mapping file, Alis Plots for identified outlets)



Result Matching_Client_Country (Document containing first attempt to match chargers with client & country)

What is needed for training, log retrieval and analysis

Scripts (Scripts for model training, Log retrieval & formatting and Analyzing Logfiles with pretrained models)



Training models, retrieving logs and analyzing data

The green marked folders hold the data needed to run the pretrained models, retrain & export models with changed parameters or retrieve logs for specific Queries and timeframes.

The following slides highlight what is needed for each of the possible usages of the set up.



Cable Predictive Maintenance Proposal (Main Documentation and manual)



Models
(Performance overview and saved models from past runs with different parameters (assessment in Performance overview))



Logs
(SeccSession and PLE dataset for Logs from August 2024, cut off at different times)



Backup



Full Report Script and Docs
(Long script for the full report, newest Parquet file and newest Masterfile)



Output (Specific Logs for Alis model, Engie mapping file, Alis Plots for identified outlets)



Result Matching_Client_Country (Document containing first attempt to match chargers with client & country)

Predictive Maintenance

Running the integrated script for the newest predictions

Scripts (Scripts for model training, Log retrieval & formatting and Analyzing Logfiles with pretrained models)



Running the reporting script

For running the reporting script, follow the manual presented in the next chapter. You need your AWS Token Vending Machine keys, run the script that is saved in the Full Report Script and Docs (works in Google Colab) and have the Parquet file and Masterdata Excel ready that is also saved there and one of the saved models from the Models folder (RNN or XGBoost).



Cable Predictive Maintenance Proposal (Main Documentation and manual)



Models (Performance overview and saved models from past runs with different parameters (assessment in Performance overview))



Logs (SeccSession and PLE dataset for Logs from August 2024, cut off at different times)



Backup



Full Report Script and Docs (Long script for the full report, newest Parquet file and newest Masterfile)



Output (Specific Logs for Alis model, Engie mapping file, Alis Plots for identified outlets)



Result Matching_Client_Country (Document containing first attempt to match chargers with client & country)

Predictive Maintenance

Retrieving Logs

Scripts (Scripts for model training, Log retrieval & formatting and Analyzing Logfiles with pretrained models)



Retrieving Logs from AWS Cloudwatch

For downloading logs, combining log datasets or cropping logs, you can find scripts in the Models/Scripts folder. Those scripts usually require your AWS access tokens as well.



Cable Predictive Maintenance Proposal (Main Documentation and manual)



Models
(Performance overview and saved models from past runs with different parameters (assessment in Performance overview))



Logs
(SeccSession and PLE dataset for Logs from August 2024, cut off at different times)



Backup



Full Report Script and Docs
(Long script for the full report, newest Parquet file and newest Masterfile)



Output (Specific Logs for Alis model, Engie mapping file, Alis Plots for identified outlets)



Result Matching_Client_Country (Document containing first attempt to match chargers with client & country)

Predictive Maintenance

Training & Finetuning the models

Scripts (Scripts for model training, Log retrieval & formatting and Analyzing Logfiles with pretrained models)



Model retraining

For retraining of models, you can also find scripts in the Models/Scripts folder. Here you have scripts where you can adjust parameters to improve your outcomes while also being able to run and test existing models. For both you require a log-based dataset, which you can either be download using your AWS credentials or you can use pre-created SeccSessionStop datasets saved in the Logs folder.



Cable Predictive Maintenance Proposal (Main Documentation and manual)



Models (Performance overview and saved models from past runs with different parameters (assessment in Performance overview))



Logs (SeccSession and PLE dataset for Logs from August 2024, cut off at different times)



Backup



Full Report Script and Docs (Long script for the full report, newest Parquet file and newest Masterfile)



Output (Specific Logs for Alis model, Engie mapping file, Alis Plots for identified outlets)



Result Matching_Client_Country (Document containing first attempt to match chargers with client & country)

Predictive Maintenance

Verify that you have all documents and information needed (run the Colab script found in Full Report and Script Docs)

Create and copy AWS tokens

Token vending machine

Search by a keyword or by column

Rows 10

Account ID	Role	Project name
689117314700	E_Developer	SiCharge D PROD
689117314700	E_Observer	SiCharge D PROD

Option 2: Setting AWS environment variables
Paste the following text in the command prompt to set the AWS environment variables.

MacOS and Linux Windows CMD Windows Powershell Settings

AWS_ACCESS_KEY_ID
ASIA2A4LUXX2GFEIMSGYO [Copy](#)

AWS_SECRET_ACCESS_KEY
4IQNCIAcUc1vD6wraPm34IqawUv05TFz30q4Pa00 [Copy](#)

AWS_SESSION_TOKEN
IQoJb3JpZ2luZVJlASaDGV1LWNlbnRyYWwMSIHMEUCIAzG6zjmQxP2FHUoMOK9Tesv+wwZ1DqFBA [Copy](#)

[Close](#)

Secrets

Notebook-Zugriff

Name	Wert	Aktionen
AWS_ACC	ASIA2A4L	🔍 📄 🗑️
AWS_SEC	nK9sA/9+	🔍 📄 🗑️
AWS_SES!	IQoJb3Jp	🔍 📄 🗑️

Newest Masterfile and Parquet file

AITeam > Predictive Maintenance > Logs

AITeam > Predictive Maintenance > Full Report Script and Docs

Masterfile



Needs to be uploaded twice

Logs



Model data

AITeam > Predictive Maintenance > Models > RNN > 4

- The model can be freely selected (the proposed model works fairly well)
- Model training can be carried out in another one of the provided scripts
- Both RNN and XGBoost can be used in this script
- If a better model is created, the report can be used just like now
- You need the config, metadata and for RNN also H5 file for the model upload

AWS Cloudwatch integration

- Go to your AWS Token Vending Machine
- Click on the Error on the left hand side
- Then go to the three dots on the very right
- Click on Settings
- Here you find the access key ID, secret Access key and session token
- Copy those keys to your secrets in Google Colab (can be found on the left sidebar with a key symbol)
- Make sure all three variables are enabled for the Notebook and paste your copied TVM values here

 689117314700 E_Observer SiCharge D PROD

SiCharge D PROD aws-001209   

Option 2: Setting AWS environment variables

Paste the following text in the command prompt to set the AWS environment variables.

MacOS and Linux

Windows CMD

Windows Powershell

Settings

MacOS and Linux Windows CMD Windows Powershell Settings

AWS_ACCESS_KEY_ID 1

ASIA2A4UXX2GP3F334E2 Copy

AWS_SECRET_ACCESS_KEY 2

zhOwMtATtKkU02gBiJNDNPyNdZRxfy5lu1B/RcsT Copy

AWS_SESSION_TOKEN 3

IQoJb3JpZ2luX2VjEiX////////wEaDGV1LWNlbnRyYWwtMSJHMEUCIQ351VAihW23C9ko+x34DI0oBE\ Copy

Secrets

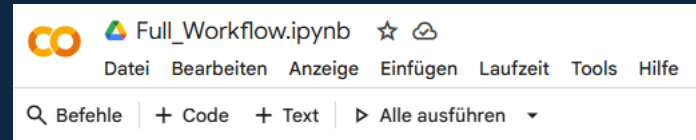
 Konfigurieren Sie Ihren Code, indem Sie Umgebungsvariablen, Dateipfade oder Schlüssel speichern. Die hier gespeicherten Werte sind privat und nur für Sie und auf von Ihnen ausgewählten Notebooks sichtbar.

 Der geheime Name darf keine Leerzeichen enthalten.

Notebook-Zugriff	Name	Wert	Aktionen
1 	AWS_ACCf	ASIA2A4U	  
2 	AWS_SECF	s8aNx4Qc	  
3 	AWS_SESS	IQoJb3Jp.	  

First upload and time range determination for AWS Query

- Now you can run your script with the newly added keys
- Make sure to have your files ready for running the script ([see here](#))
- The script is now asking you to upload the newest Master file for the outlet failures
- Upload the file (this should only take a couple of seconds)
- Upon finishing, the model script will give you the summary of what you uploaded **1**
- Now, the newest timestamp from your file is identified, your methods run automatically and the range for the download of new logs identified
- The AWS Query name and other variables are already hard coded into your script



1 Please select your master Excel file containing all past analyzed issues from your local machine to upload:
Dateien auswählen Mastero...2230.xlsx

First 5 rows of the loaded DataFrame:

	Outlet ID	ID	Outlet	Detection Date	Counter
0	0AQbZI1	0AQbZI	1	2025-09-09 09:58	1
1	0PrWyp3	0PrWyp	3	2025-09-07 13:24	1
2	17xs0j2	17xs0j	2	2025-09-07 10:55	1
3	198r072	198r07	2	2025-09-05 01:21	1
4	198r074	198r07	4	2025-08-27 11:45	1

DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 380 entries, 0 to 379
Data columns (total 5 columns):
Column Non-Null Count Dtype

0 Outlet ID 380 non-null object
1 ID 380 non-null object
2 Outlet 380 non-null int64
3 Detection Date 380 non-null object
4 Counter 380 non-null int64
dtypes: int64(2), object(3)
memory usage: 15.0+ KB

AWS Log download

- Now, the first execution block is executed
- If run sporadically, the script automatically chunks the time (creates subsets) to save costs in AWS
- This means that 2-day time windows are run for log retrieval
- The model is designed to run the queries in a looped and decreasing timeframe to ensure all logs are recovered
- For up to three days, the model downloads all new SessSessionStop logs up to date (after your last timestamp identified in your Excel Masterfile)

```
Start : 2025-09-09 00:00:00+00:00
End   : 2025-09-10 23:59:59+00:00
Short time period (1 days). Using optimized single-query approach.
New credentials obtained. Expiration: 2025-09-10 11:42:04+00:00
Changed sort order from desc to asc for proper pagination
Starting batch with window 2025-09-09T00:00:00+00:00 → 2025-09-10T23:59:59+00:00 (limit 10000)
Batch status: Complete, rows: 10000
Collected this round: 10000, total collected: 10000
Starting batch with window 2025-09-09T11:32:25.880000+00:00 → 2025-09-10T23:59:59+00:00 (limit 10000)
Batch status: Complete, rows: 10000
Collected this round: 10000, total collected: 20000
Starting batch with window 2025-09-09T17:12:34.078000+00:00 → 2025-09-10T23:59:59+00:00 (limit 10000)
Batch status: Complete, rows: 10000
Collected this round: 10000, total collected: 30000
Starting batch with window 2025-09-10T05:27:51.937000+00:00 → 2025-09-10T23:59:59+00:00 (limit 10000)
Batch status: Complete, rows: 7221
Collected this round: 7221, total collected: 37221
Saved all 37221 rows to single files
```

	IDOutlet	@timestamp	diff	Energydiff	PLE_Error	\
0	otaDfB2	2025-09-09 00:00:05.697	0.7774	0.01968	false	
1	otaDfB3	2025-09-09 00:00:07.275	0	0	false	
2	otaDfB2	2025-09-09 00:00:07.757	1.2957	0.05688	false	
3	otaDfB3	2025-09-09 00:00:07.758	6.7375	0.2432	false	
4	otaDfB1	2025-09-09 00:00:08.402	2.0731	0.09939	false	

```
@ptr
0 CpABck8KFTY40TExNzMxNDcwMDptZXNzYWdlcXACGiQ5ZD...
1 CpABck8KFTY40TExNzMxNDcwMDptZXNzYWdlcXAAGiQ5ZD...
2 CpABck8KFTY40TExNzMxNDcwMDptZXNzYWdlcXAFGiQ5ZD...
3 CpABck8KFTY40TExNzMxNDcwMDptZXNzYWdlcXAFGiQ5ZD...
4 CpABck8KFTY40TExNzMxNDcwMDptZXNzYWdlcXABGiQ5ZD...
```

Step 1

Creating the new dataset for problem identification

- Now you need to upload the newest Parquet file available
- This file contains the logs of the 100 days before the script was last run
- The upload might take a bit of time, afterwards there is a short summary on the uploaded file
- Some preprocessing is carried out
- In the next step, the old 100-day dataset is combined with the newly retrieved logs, duplicates are removed (if there is an overlap) and the file is formatted to once again contain the logs for 100 days
- The download of the new Parquet file is triggered (but will only be carried out once the whole script is run)
- Please replace the Parquet file in the folder used after running the script

Please select the Parquet file from your local machine to upload:

Dateien auswählen newest_...2.parquet

First 5 rows of df_new:

	IDOutlet	datetime	diff	Energydiff	PLE_Error	\
0	otaDfB2	2025-09-09 00:00:05.697	0.7774	0.01968	false	
1	otaDfB3	2025-09-09 00:00:07.275	0	0	false	
2	otaDfB2	2025-09-09 00:00:07.757	1.2957	0.05688	false	
3	otaDfB3	2025-09-09 00:00:07.758	6.7375	0.2432	false	
4	otaDfB1	2025-09-09 00:00:08.402	2.0731	0.09939	false	

@ptr

0	CpABck8KFTY40TExnzMxNDcwMDptZXNzYwdlcxACGiQ5ZD...
1	CpABck8KFTY40TExnzMxNDcwMDptZXNzYwdlcxAAGiQ5ZD...
2	CpABck8KFTY40TExnzMxNDcwMDptZXNzYwdlcxAFGiQ5ZD...
3	CpABck8KFTY40TExnzMxNDcwMDptZXNzYwdlcxAFGiQ5ZD...
4	CpABck8KFTY40TExnzMxNDcwMDptZXNzYwdlcxABGiQ5ZD...

```
Combined shape before de-dup: (2722728, 6)
Shape after de-dup: (2711302, 6)
CUTOFF timestamp: 2025-06-02 11:46:59.784229+00:00
Datetime column timezone: UTC
Shape after CUTOFF filter: (2697941, 6)
Final DataFrame shape: (2697941, 6)
Final DataFrame columns: ['IDOutlet', 'datetime', 'diff', 'Energydiff', 'PLE_Error', '@ptr']
File saved as: newest_cable_temp_Logs_20250910_114700.parquet
```

Uploading the Masterfile and selecting a model type

- Now the model runs a long part of the script containing all functions we need for the main execution (no input needed)
- Next, the main function is run and executed
- First you receive a quick summary and some notes
- Next, the model asks you to re-upload the Masterfile (same as in step 2)
- Afterward, the script asks you to insert a number to select the model you want to integrate (select one for RNN and two for XGBoost)
- Models can be found as [described here](#)

```
🔗 USAGE OPTIONS:

🔗 AUTOMATED WORKFLOW:
• All file uploads happen at the beginning
• Script runs automatically without user intervention
• Complete end-to-end analysis pipeline

📄 WORKFLOW STEPS:
1. Upload all required files
2. Automatic data collection and preprocessing
3. Model inference and analysis
4. Masteroutlet file integration (if provided)
5. Comprehensive outlet list creation
6. Optional CloudWatch queries with filtering

🔗 Re-Upload the Masterfile for Outlet issues...
Dateien auswählen Mastero...2230.xlsx

🔗 UPLOADING MODEL FILES...
=== LOADING UPLOADED MODEL FILES ===

📄 MODEL TYPE SELECTION:
1. RNN Model (.h5, .hdf5, .keras)
2. XGBoost Model (.json)
3. Auto-detect (recommended)
Choose model type (1-3, default=3): 1
🔗 Selected: RNN Model
🔗 Please upload your RNN model file (.h5, .hdf5, .keras):
```

Model upload and pre-processing

- Now, you need to upload your model file (h5 for RNN, json for XGBoost) **1**
- Then, you upload the metadata file **2**
- And finally, the feature config file needs to be uploaded **3**
- Now, all uploads are finished, and the script runs without needing any more input
- This step usually takes a longer amount of time
- You first get a summary on the model you uploaded
- Followed from a step where all data needed is loaded and preprocessed (details are shown in the output of the script)

1 Selected: RNN Model
Please upload your RNN model file (.h5, .hdf5, .keras):
Dateien auswählen rnn_model(1).h5

2 Please upload your metadata file (.json):
Dateien auswählen rnn_mod...ata.json

3 Please upload your feature config file (.json) - optional:
Dateien auswählen rnn_feat...onfig.json

Model: "sequential"

Layer (type)	Output Shape	Param #
simple_rnn (SimpleRNN)	(None, 1, 32)	1,312
dropout (Dropout)	(None, 1, 32)	0
simple_rnn_1 (SimpleRNN)	(None, 16)	784
dropout_1 (Dropout)	(None, 16)	0
dense (Dense)	(None, 16)	272
dropout_2 (Dropout)	(None, 16)	0
dense_1 (Dense)	(None, 1)	17

Total params: 2,387 (9.33 KB)
Trainable params: 2,385 (9.32 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 2 (12.00 B)

=====

📁 STEP 2: DATA LOADING AND PREPROCESSING

=====

Failure predictions

- In step 3 of the script, the analysis of the newly created Parquet dataset is carried out and failure predictions created
- You will see an overview, followed by a long table showing all events, where a temperature difference spike between the sensors was detected
- This is followed by a longer list of all outlet failure predictions with its summary
- Finally, the results are aggregated in an Excel report
- The download is also triggered directly and carried out after the script is fully run

Generating analysis reports...

=== GENERATING FREAK EVENTS AND WARNINGS (RNN) ===

LAST 30 DAYS ANALYSIS:

- Total records in last 30 days: 823,262
- Unique outlets in last 30 days: 5,155

FREAK EVENTS DETECTION (Last 30 Days):

- Detection criteria: High prediction probability + extreme values

FREAK EVENTS TABLE (Last 30 Days):

Outlet ID	Freak Severity	Detection Date	Risk Level
bwnWTj3	100.0%	2025-09-10 10:44	CRITICAL
obYNIH1	100.0%	2025-09-10 11:11	CRITICAL
RFhsW11	100.0%	2025-09-10 10:58	CRITICAL
k90ExG2	100.0%	2025-09-10 09:26	CRITICAL
XiHDgt1	100.0%	2025-09-04 12:36	CRITICAL

FREAK EVENTS RISK BREAKDOWN (Last 30 Days):

- Critical ($\geq 95\%$): 77 outlets
- High (90-94%): 16 outlets
- Medium (80-89%): 10 outlets

PROBLEM PREDICTION CATEGORIES (Last 30 Days):

Category	Outlet ID	Prediction %	Detection Date	Action Required
Critical Risk	U6H3Gj2	97.9%	2025-08-11 13:18	IMMEDIATE
Critical Risk	HvwKnz1	97.8%	2025-08-11 22:44	IMMEDIATE
Critical Risk	wEivGK2	97.3%	2025-08-12 08:14	IMMEDIATE
Critical Risk	hXcXbF1	97.7%	2025-08-12 15:43	IMMEDIATE
Critical Risk	TmguWp1	97.7%	2025-08-14 00:14	IMMEDIATE
Critical Risk	acno331	98.1%	2025-08-14 07:00	IMMEDIATE
Critical Risk	acno334	98.9%	2025-08-14 07:04	IMMEDIATE
Critical Risk	CJIvv01	98.8%	2025-08-14 07:52	IMMEDIATE
Critical Risk	MCZnP31	97.3%	2025-08-15 09:37	IMMEDIATE
Critical Risk	gu1dvM3	97.9%	2025-08-16 09:32	IMMEDIATE

PROBLEM PREDICTION CATEGORY STATISTICS (Last 30 Days):

- Critical Risk ($\geq 90\%$): 95 outlets
- High Risk (80-89%): 22 outlets
- Medium Risk (60-79%): 291 outlets
- Total Problem Predictions: 408 outlets

SUMMARY:

- Freak Events Detected: 103 unique outlets
- Problem Predictions: 408 unique outlets
- Total Outlets Analyzed: 5155
- Analysis Period: Last 30 days

=== CREATING COMPREHENSIVE EXCEL REPORT ===

Creating Freak Events sheet...

Creating Risks sheet...

Creating Summary sheet...

Excel report created: RNN_Analysis_Report_20250910_120102.xlsx

- Sheet 1: Freak Events (103 entries)
- Sheet 2: Risk Categories (408 entries)
- Sheet 3: Summary

File downloaded: RNN_Analysis_Report_20250910_120102.xlsx

Updating the Masterfile

- In step 4, the Masterdata file is updated
- This is the file containing all outlets identified in the past
- First, the script compares the new results to the existing Excel file
- In case of outlets already identified, the counter adds one to each Outlet in the file
- If the identification timestamp is more recent, it is overwritten with the newer time
- If additional outlets are identified, they get appended to the list while also being added to the newly identified sheet of the Excel file
- Finally, the sheet showcasing the 50 most recently identified outlet issues is updated with the new data

```
✓ All required sheets loaded/created
Master_List loaded with 380 existing entries
Updating existing entry for outlet: U6H3Gj2
Updated counter: 1 → 2
```

```
Updating existing entry for outlet: ev8HNP2
Updated detection date: 2025-09-08 12:59 → 2025-09-09 17:35
Updated counter: 1 → 2
Updating existing entry for outlet: oLG0BP1
Updated detection date: 2025-09-09 10:10 → 2025-09-09 19:00
Updated counter: 1 → 2
Updating existing entry for outlet: oLG0BP3
Updated detection date: 2025-08-29 13:36 → 2025-09-09 19:49
Updated counter: 1 → 2
Updating existing entry for outlet: 5R9akG2
Updated detection date: 2025-09-09 09:46 → 2025-09-09 19:53
Updated counter: 1 → 2
```

```
Integration Summary:
• Updated existing entries: 408
• Added new entries: 0
```

```
Updating 50_newest_timestamps sheet...
50_newest_timestamps sheet updated with 50 entries
Updated Masteroutlet file saved as: Masteroutlet_File_Updated_20250910_120104.xlsx
File downloaded: Masteroutlet_File_Updated_20250910_120104.xlsx
Integration completed successfully!
Updated file: Masteroutlet_File_Updated_20250910_120104.xlsx
```

Creation of the comprehensive outlet list

- In step 5, a list of the most relevant outlets is compiled
- This list is used to implement a filter for the next step
- All newly identified outlets, the 50 most recent outlets identified and the top 10 in terms of errors counted is put together
- Duplicates of course are removed
- To be able to look at outlets individually, a value called OutletID is used, which is comprised of charger ID and outlet number
- The comprehensive list is saved and ready to be used in step 6

```
=== CREATING COMPREHENSIVE OUTLET LIST ===  
📁 Loading Masteroutlet file: Masteroutlet_File_Updated_20250910_120104.xlsx  
✅ All required sheets loaded  
📊 Processing newest 50 timestamps...  
✅ Added 50 newest timestamp entries  
📊 Processing newly identified outlets...  
✅ Added 111 newly identified entries  
📊 Processing top 10 counted outlets...  
✅ Added 5 top counted entries
```

```
📊 COMPREHENSIVE LIST SUMMARY:  
• Total unique outlets: 166  
• Newest timestamps: 50  
• Newly identified: 50  
• Top counted: 5  
• Duplicates removed: -61
```

```
📁 Comprehensive outlet list exported to: Comprehensive_Outlet_List_20250910_120104.xlsx  
📁 Exported to: Comprehensive_Outlet_List_20250910_120104.xlsx  
🌐 Comprehensive outlet list stored in global variable 'comprehensive_outlet_list'
```

Creation of specific log files for visualization script

- In step 6, three additional Cloudwatch queries are carried out
- The logic is similar to the first download, only this time, there is always a custom chunking carried out (varies per query)
- In total three months worth of data are downloaded for the OutletIDs identified in Step 5 (direct filtering)
- The queries are in relation to PLE, CTD and SeccSessionstop Logs and used in the visualization script
- You can interrupt the script if this step should be skipped (quite costly in AWS)
- If run completely, 3 separate CSV files are created and downloaded
- **Please update the Master Excel file and Parquet file that are now downloaded in the folder!**

```
🚀 RUNNING AUTOMATED CLOUDWATCH QUERIES WITH FILTERING
Query names: ['Niklas/Pred_CTD', 'Niklas/Pred_PLE', 'Niklas/Pred_SeccSessStop']
📅 Time range: 2025-06-12 to 2025-09-10
📦 Chunk size: 7 days per chunk
⚠️ Make sure these saved queries exist in your CloudWatch Logs Insights
```

```
=== RUNNING CHUNKED FILTERED CLOUDWATCH QUERIES ===
📅 Time range: 2025-06-12 to 2025-09-10
🔍 Using comprehensive outlet list with 166 outlets
📊 Filter criteria:
• Outlet IDs: 166 unique values
• IDs: 150 unique values
```

```
🔍 QUERY 1: Niklas/Pred_CTD
📄 Description: CTD notification Query with CableTempSensor difference thresholds
Using 3-day chunks for this query
Using 3-day chunks for this query
```

```
=== Processing Chunk 1: 2025-06-12 to 2025-06-15 ===
🚀 Executing CloudWatch query for chunk 1...
New credentials obtained. Expiration: 2025-09-10 12:16:06+00:00
Added sort by @timestamp desc (default for CloudWatch)
Starting batch 1 with window 2025-06-12T12:01:04.502131+00:00 → 2025-06-15T12:01:04.502131+00:00 (limit 10000)
Batch 1 status: Complete, rows: 9553
Processed 9553 rows from CloudWatch. Total unique collected so far: 9553
📄 Next batch will search up to: 2025-06-12T12:01:04.606999+00:00
Fewer than 10000 results returned (9553). Likely end of data for this time window. Stopping pagination.
✅ Pagination complete: 9553 total unique results across 1 batches
✅ Query executed successfully. Raw results: 9553 rows
• Results already in correct format: 9553 rows
• Column names: ['@timestamp', 'IDOutlet', 'cause', 'cableid', 'nowcur', 'Contacttemp', 'Contacttemp2', 'diff', 'factor', 'EVID', 'Energy', 'duration', '@ptr']
📊 Raw data shape: (9553, 13)
Applying PLE Error specific filtering...
• Available columns: ['@timestamp', 'IDOutlet', 'cause', 'cableid', 'nowcur', 'Contacttemp', 'Contacttemp2', 'diff', 'factor', 'EVID', 'Energy', 'duration', '@ptr']
• DataFrame shape: (9553, 13)
• Found IDOutlet column - using direct filtering
• IDOutlet matches: 1,638 rows
• Final filtered: 1,638 rows
• Reduction: 82.9%
✅ Chunk 1 filtering completed:
• Raw rows: 9553
• Filtered rows: 1638
• Filter stats: {'idoutlet_match': 1638, 'total_rows': 9553, 'filtered_rows': 1638, 'reduction_percentage': 82.85355385742699}

=== Processing Chunk 2: 2025-06-15 to 2025-06-18 ===
🚀 Executing CloudWatch query for chunk 2...
New credentials obtained. Expiration: 2025-09-10 12:16:31+00:00
Added sort by @timestamp desc (default for CloudWatch)
Starting batch 1 with window 2025-06-15T12:01:04.502131+00:00 → 2025-06-18T12:01:04.502131+00:00 (limit 10000)
```

What does the script do?

Update Masterdata Excel

- Needs to be uploaded, then is edited in the script
- New outlets identified are gathered
- Counter for outlets continuously identified and 50 newest outlets identified report

Download newest training data and report

- Update the Parquet file with all Outlets (100-day window from day of running)
- Can be used for training new models
- Used for analyzing new failures in the script directly

Download identified outlets logs for Alis visuals

- Automatic CSV download of affected outlets
- Three-month data for PLE, CTD and SeccSessStop
- Pre filtered & formatted for direct input into Alis model

Further remarks

Please save the newest data in the folder to be up to date for running the script

AI Team > Predictive Maintenance > Full Report Script and Docs 

Feel free to test out other models for the script

Running a script biweekly should be a good timeframe

Running the script too often makes the counter function in the Master Excel obsolete

Feel free to change the downloaded Parquet and CSV timeframes (e.g. longer visuals for the KPIs). They are easily found in the script

A lot is Hardcoded in the script and can be found by scrolling through "Main"